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Process Crystallographic Simulation for Biocompatible Piezoelectric Material Design and Generation

Eiji Nakamachi · Yasutomo Uetsuji · Hiroyuki Kuramae · Kazuyoshi Tsuchiya · Hwisim Hwang

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Abstract From 1880's discovery of piezoelectricity by French physicists Jacques and Pierre Curie, a huge number of piezoelectric materials have been developed and applied to the industrial equipment and scientific instrument. In the middle of 20th century, most widely used ceramic piezoelectric materials, $BaTiO_3$ (BTO) in 1944, $PbTiO_3$ in 1950, and $Pb(Zr,Ti)O_3$ (PZT) in 1955, were discovered through "hermetic art approach." It was not CAE driven material discovery. Actually, the experimental trial and error approach is inefficient way for the material discovery. Therefore a new CAE technique to develop a new high performance piezoelectric material under a short lead time is strongly required. It can analyze material characteristics, and design material structure and generation process simultaneously before actual production. This overall CAE technique for new material design and generation, which can be called as "process crystallographic simulation," is discussed in this state-of-the-art paper, which will be able to establish a new concept of material and process design.

Now, we have serious problem with the piezoelectric material. Actually, PZT is most used in the world. However, "lead," which is a component of PZT-based piezoelectric material, is the toxic material. The usage of lead and toxic materials is prohibited by the waste electrical and electronic equipment (WEEE) and the restriction on hazardous substances (RoHS). Therefore, CAE driven new biocompatible material development is recognized as urgent subject. A goal is to develop an environmentally and biologically compatible piezoelectric material, which can be applied for human healthcare devices, such as Bio-MEMS devices. Until now, we have CAE methodologies to develop a new material, such as the atomic simulation, the continuum mechanics base finite element method, and the crystal process optimization method, but these are not cooperated effectively. An overall and simultaneous computational technique is strongly required. In this review paper, we survey and discuss numerical methodologies, "process crystallographic simulation," for material and generation process design. Further, an invention of a new biocompatible piezoelectric material, its generation and validation of a newly developed numerical technique, are demonstrated.

In this paper, below described subjects are reviewed and discussed.

1. Numerical analysis technique, "process crystallographic simulation," which consists of a three-scale structure analysis and a generation process analysis.
2. Material and Process design of new biocompatible piezoelectric materials.

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3. Generation of $MgSiO_3$ thin film by using radio-frequency (RF) magnetron sputtering system. Validation of CAE technique.

Consequently, a general concept of CAE driven material discovery technique could be understood through this state-of-the-art paper.

1 Introduction

We recognized that CAE technique for a new material and generation process design are "now a day subject" in the aspect of application in the industry. In this review paper, we select a particular material, which should be developed immediately for the world scientific and engineering requirements, and discuss CAE technique to solve this industrial needs.

(A) Material Until now, lead zirconate titanate (Pb(Zr, Ti)O_3:PZT) has been used most widely for sensors [1–6], actuators [7–10], memory devices [11–13] and micro electro mechanical systems (MEMS) [14–20], because of its high piezoelectric and dielectric properties. The piezoelectric thin film with aligned crystallographic orientation shows the highest piezoelectric property than any polycrystalline materials with random orientations. Sputtering [21–26], chemical or physical vapor deposition (CVD or PVD) [27–32], pulsed laser deposition (PLD) [33–38] and molecular beam epitaxy (MBE) [39–42] are commonly used to generate high performance piezoelectric thin films. Lattice parameters and crystallographic orientations of epitaxially grown thin films on various substrates can be controlled by these procedures. H. Funakubo et al. [43] generated [001] and [100]-orientated PZT thin films on MgO (001) substrate by using CVD method. They succeeded to obtain a huge strain caused by the two effect: the synergetic effect of [001] orientation with the piezoelectric strain; and the strain effect of [100] orientation caused by switching under conditions of the external electric field. Additionally, PZT-based piezoelectric materials, such as $Pb(Zn_{1/3}Nb_{2/3})O_3$–$PbTiO_3$ [44] and $PbMg_{1/3}Nb_{2/3}O_3$–$PbTiO_3$ [45], have also been developed [46–48].

However, "lead," which is a component of PZT-based piezoelectric material, is the toxic material. The usage of lead and toxic materials is prohibited by the waste electrical and electronic equipment (WEEE) [49] and the restriction on hazardous substances (RoHS) [50].

For alternative piezoelectric materials of the PZT, lead-free piezoelectric materials have been studied. Zhu et al. [51] generated [111]-orientated $BaTiO_3$ on $LaNiO_3$(111) substrate, which had a crystallographic orientation with maximum piezoelectric strain constants. Zhang et al. [52] doped Ca and Zr in $BaTiO_3$ (BTO) and succeeded in generating the piezoelectric material with high piezoelectric properties. Further, Fu et al. [53] doped La_2O_3 in Bi-based $(Bi_{0.5}Na_{0.5})_{0.94}Ba_{0.06}TiO_3$ and succeeded in generating a high performance piezoelectric material. However, their goals were to develop an environmentally compatible piezoelectric material, and the biocompatibility of their piezoelectric materials has not been investigated. Therefore, their piezoelectric materials could not be applied for human healthcare devices, such as Bio-MEMS devices.

Bio-MEMS, which can be applied to the health monitoring system and the drug delivery system, is one of most attractive subject in the research field of nano- and biotechnology. However, they have many difficulties to design new biocompatible materials and find optimum generation process.

(B) CAE Technique In this state-of-the-art paper, the numerical technologies of analysis and design for a new material, and optimize a generation process are reviewed. This numerical technology can be categorized as "process crystallographic simulation technology." This CAE driven material and device discovery is strongly required, due to time consuming experimental approach such as finding an optimum sputtering process for new materials and a substrate crystal structure through enormous experimental trials. This scheme could find an optimum crystal structure for a high piezoelectric property, and an optimum generation process for a high quality thin film.

Until now, the conventional analysis schemes, such as the molecular dynamics (MD) method [54–58] and the first-principles calculation based on the density functional theory (DFT) [59–64], have been applied to the crystal growth process simulations. The MD method has been used mainly to analyze the crystal growth process of pure atoms. Lee et al. [65] analyzed Ni growth on Al surface and Xu et al. [66] calculated the Ge growth on Si(111). In the cases of the perovskite compounds, the MD method has been applied to analyze the phase transition, the polarization switching and properties of crystal depending on temperature and pressure. Paul et al. [67] analyzed the phase transition of $BaTiO_3$ caused by rising temperature and Costa et al. [68] analyzed the one of $PbTiO_3$ caused by rising temperature and pressure. Sepliarsky et al. [69] calculated the polarization switching of the $KNbO_3$ and Zhang et al. [70] analyzed the elastic moduli of the $CaSiO_3$ caused by rising temperature and pressure. However, the reliability of its numerical results is poor due to its uncertain inter-atomic potentials for the various combinations of atoms. The MD method could not predict the differences of poly-crystal structures and material properties caused by changing combinations of the crystals and the substrates. It can be concluded that the

conventional MD method has many problems for the crystal growth prediction of perovskite compounds grown on the arbitrarily selected substrates.

On the other hand, the DFT can treat interactions between electrons and protons, therefore the reliable inter-atomic potentials can be obtained. The first-principles calculations based on the DFT were applied to the epitaxial growth of the ferroelectric material by Diegueaz et al. [71] and Yakovkin et al. [72]. Diegueaz et al. evaluated the stress increase and the polarization change caused by the lattice mismatch between a substrate and a thin film crystal, such as $BaTiO_3$ and $PbTiO_3$ [71]. Similarly, Yakovkin et al. has investigated in the case of $SrTiO_3$ thin film growth on Si substrate [72]. However, these analyses adopted limited assumptions, such as fixing the conformations of thin film crystals and the growth orientations on the substrates. In this conventional algorithm, the grown orientation is determined by the purely geometrical lattice mismatch between thin films and substrates. This algorithm can be concluded as the simple and effective scheme to predict the preferred orientation of the thin film.

(C) Material Generation and CAE Technique Validation In order to generate the new piezoelectric thin film, a crystal growth process of the thin film should be predicted accurately. The stable crystal cluster of the thin film, which consists geometrically with substrate crystal, should be grown on the substrate. Generally, the crystal cluster is an aggregate of thin film crystals. Their morphology and orientations were varied according to the combination of the thin film and the substrate crystals. Therefore, the numerical analysis scheme of the crystal growth process, which can find the best combination of the thin film and the substrate crystal, is strongly required, to optimize the new piezoelectric thin film.

In this paper, following subjects are discussed to develop a new biocompatible piezoelectric material $MgSiO_3$, which is found as a top candidate, by using process-crystallographic simulation, and generate thin films and validate the numerical simulation scheme by observing the crystal structure and piezoelectricity.

1. Development of a numerical analysis technique of a three-scale structure analysis and a process-crystallographic simulation.
2. Design of new biocompatible piezoelectric materials. It has the perovskite oxide component and the tetragonal crystal structure.
3. Generation of $MgSiO_3$ biocompatible piezoelectric thin films by using radio-frequency (RF) magnetron sputtering system, where an optimum generating condition is found by using the experimental design algorithm and the response surface method.
4. Validations of numerical technique through measurements of piezoelectricity and biocompatibility of discovered piezoelectric material $MgSiO_3$.

2 Three-Scale Computational Methodology

It is necessary for CAE driven material discovery and process design to consider three scales which are crystal structure, microstructure and macrostructure. Crystal structure is an origin of excellent material physicality. Microstructure, which indicates crystal morphology, has a strong influence on material property. Macrostructure dominates the performance of mechanical electro device and system. This chapter review CAE simulation technique coupling the above three scales, which consists of the physical and mathematical modeling to characterize and design epitaxially grown new lead-free piezoelectric thin films.

Table 1 shows the comparison of typical characteristics of typical CAE simulation methods, which are the first-principles calculation based on density functional theory (DFT), molecular dynamics (MD) simulation and finite element method (FEM) simulation based on continuum mechanics. DFT calculations can solve static problem of atoms and molecules. Although there are some challenges for finite temperature problem [73], DFT calculations generally ignore atomic vibration assuming absolute zero state. Even if there is no experimental data, we can understand total energy, stable crystal structure and material property through DFT calculations. On the other hand, MD simulation enable us evaluate dynamic behaviors of atoms, crystal structures and microstructures if the function of potential energy can be constructed by experimental data or the first-principle calculations. Additionally FEM is a well-known CAE simulation method for performance evaluation of devise and system. As FEM can treat various static and dynamic problems at finite temperature, it is widely applied to various products. These CAE simulation methods have some advantages and disadvantages, and the scale of their fields is different. Consequently, as new biocompatible lead-free piezoelectric material discovery and process design demands computational simulation in the wide range from crystal structure to microstructure and macrostructure, it is necessary to combine these CAE simulations and to connect material property and behavior at different scales.

In this study, Fig. 1 shows the schematic description of the three-scale modeling of a new piezoelectric thin film, which is grown on a substrate. It shows the three-scale structures, such as a “crystal structure”, a “micro polycrystalline structure” and a “macro continuum structure”. At the scale of crystal structure, stable crystal structures and material properties are evaluated by using the first-principles calculation. Preferred orientations and their fraction, which de-

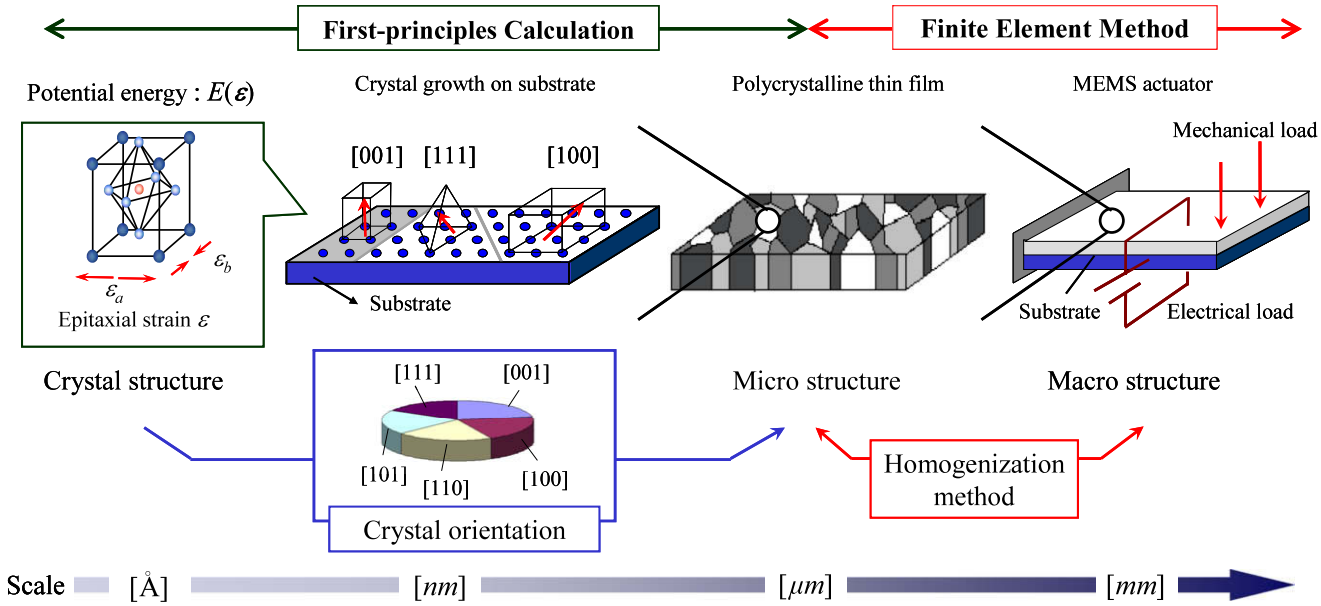


Fig. 1 Tree-scale modeling of piezoelectric thin film and flow of three-scale structure and process crystallographic analyses

Table 1 Comparison of characteristics of typical CAE simulation methods

		First-principles calculation DFT	MD	Continuum mechanics FEM
Equation		Schroedinger equation	Motion equation	Equilibrium equation
Scale	Electron	○	×	×
	Atom	○	○	×
	Crystal structure	○	○	×
	Microstructure	×	△	○
	Macrostructure	×	×	○
Field		Static	Dynamic	Static and dynamic
Temperature		Absolute zero	Finite temperature	
Objective		Structural stability Material property	Crystal morphology Material behavior	Performance of devise

termine the crystal morphology of microstructure, are calculated by using the process crystallographic simulation at the microscopic scale. The macroscopic continuum structure analysis provides the piezoelectric properties of the thin film by using the finite element analysis on basis of the crystallographic homogenization theory. Therefore, the three-scale structure analysis can predict the epitaxial growth process of not only the existent piezoelectric materials but also the new ones.

2.1 Piezoelectric Material

In general, a group of dielectric materials can be classified into four types, which are dielectrics, piezoelectrics, pyroelectrics and ferroelectrics. The dielectrics and piezoelectrics generate internal electric polarization by the external electric loads and mechanical loads, respectively. On the other hand, pyroelectrics and ferroelectrics have spontaneous polarization without the external loads. Pyroelectrics change its spontaneous polarization according to temperature changes. Additionally ferroelectrics switch the direction of spontaneous polarization by the external electric or mechanical loads. This phenomenon of ferroelectrics is called "domain switching". As the directions of spontaneous polarization are random in domains of single crystal and grains of polycrystal, the total of dipole is macroscopically zero. However, when the large external electric load is applied to ferroelectrics, the direction of spontaneous polarization

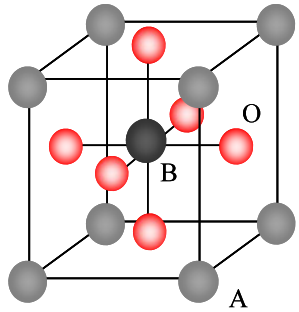


Fig. 2 Crystal structure of perovskite-type oxide ABO_3

of each domain is arranged to the intended direction owing to domain switching. And then the macroscopic dipole of ferroelectrics can be increased. The control of spontaneous polarization enables ferroelectrics produce excellent piezoelectricity. Therefore a lot of ferroelectrics are utilized as a piezoelectric material for various mechanical electro devises and systems. Especially, perovskite-type oxides as shown in Fig. 2, such as barium titanate ($BaTiO_3$), lead titanate ($PbTiO_3$) and lead zirconate titanate ($Pb(Zr,Ti)O_3$:PZT), are well-known as excellent materials.

The mechanical and electric behaviors of piezoelectric materials can be expressed by the following d-form constitutive equations.

$$\varepsilon_{ij} = s^E_{ijkl}\sigma_{kl} + d_{nij}E_n, \tag{1}$$

$$D_n = d_{nij}\sigma_{ij} + \epsilon^T_{mn}E_m, \tag{2}$$

where ε_{ij} and σ_{ij} are strain and stress tensors, D_n and E_n are the electric displacement and electric field vectors, respectively. s^E_{ijkl} is the elastic compliance tensor at constant electric field and ϵ^T_{mn} is the dielectric constant tensor at constant stress. Furthermore d_{nij} denotes the piezoelectric strain constant tensor. The first term at the right of Eq. (2) indicates the effect of mechanical stress on electric displacement, which is called "piezoelectric effect". On the other hand, The second term at the right of Eq. (1) indicates the effect of electric field on mechanical strain, which is called "inverse piezoelectric effect".

If the constitutive equations change from d-form to e-form, the piezoelectric elastic constitutive equations are described as,

$$\sigma_{ij} = C^E_{ijkl}\varepsilon_{kl} - e_{nij}E_n \tag{3}$$

$$D_n = e_{nij}\varepsilon_{ij} + \epsilon^S_{mn}E_m \tag{4}$$

where C^E_{ijkl} denotes the elastic stiffness tensor at constant electric field, and ϵ^S_{mn} is the dielectric constant tensor at constant strain. e_{nij} indicates the piezoelectric stress constant tensors. Two elastic constants and two piezoelectric constants are related to as,

$$C^E_{ij}s^E_{jk} = \delta_{ik}, \tag{5}$$

$$e_{ij} = d_{ik}C^E_{kj}, \tag{6}$$

where δ_{ik} is the Kronecker delta.

2.2 First-Principle Calculation

The Schrödinger equation in polyatomic system has been described by using Hartree and Hartree-Fock approximations. These approximations assume the wave function in polyatomic system for the Hamiltonian. On the other hand, the energy equation, which is the function of ionic density with ground state, and the electronic density function are defined. Hohenberg and Kohn [74] proved that the Schrödinger equation can be soluble in principle by using the electronic density function and constructed the density functional theory (DFT). The Schrödinger equation in N-electron system is described as,

$$\left[\sum_{i=1}^{N} -\frac{\hbar^2}{2m}\nabla^2 + \sum_{i=1}^{N} V_{ext} + \sum_{i=1}^{N} V_{eff}\right]\Psi_i(\mathbf{r}_1,\ldots,\mathbf{r}_N) = E\Psi_i(\mathbf{r}_1,\ldots,\mathbf{r}_N), \tag{7}$$

where $\hbar$ is the Planck's constant, V_{ext} the external potential, V_{eff} the effective potential, $\Psi_i(\mathbf{r}_1,\ldots,\mathbf{r}_n)$ the wave function and E the energy of system. The electronic density $n(\mathbf{r})$ is described as,

$$n(\mathbf{r}) = \sum_{i=1}^{N} \Psi_i^*(\mathbf{r})\Psi_i(\mathbf{r}). \tag{8}$$

The external field $\nu(\mathbf{r})$ and the wave function of ground state $\Psi_i(\mathbf{r}_1,\ldots,\mathbf{r}_n)$ are uniquely determined by the electronic density $n(\mathbf{r})$. The ground state energy $E[n(\mathbf{r})]$ is given by

$$E\big[n(\mathbf{r})\big] = F\big[n(\mathbf{r})\big] + \int \nu(\mathbf{r})n(\mathbf{r})d\mathbf{r}, \tag{9}$$

$$F\big[n(\mathbf{r})\big] = M_S\big[n(\mathbf{r})\big] + \frac{e^2}{2}\int \frac{n(\mathbf{r})n(\mathbf{r}')}{|\mathbf{r}-\mathbf{r}'|}d\mathbf{r}d\mathbf{r}' + E_{xc}\big[n(\mathbf{r})\big], \tag{10}$$

where $F[n(\mathbf{r})]$ is the expectation of the motional energy and the interaction energy. It is independent to the external field $\nu(\mathbf{r})$ and is uniquely indicated for the universal function by $n(\mathbf{r})$. $M_S[n(\mathbf{r})]$ is the motional energy of the fermion system without interaction and $E_{xc}[n(\mathbf{r})]$ is the exchange correlation energy.

Kohn and Sham [75] proposed the practical apporoximation method of ion state in the monoelectronic equation, which is called as the Kohn-Sham equation.

$$\left[-\frac{\hbar}{2m}\nabla^2 + V_{eff}(\mathbf{r})\right]\psi_i(\mathbf{r}) = \varepsilon_{me}\psi_i(\mathbf{r}), \tag{11}$$

where ε_{me} is the eigenvalue of the monoelectron equation. The effective potential V_{eff} is described as,

$$V_{eff}(\mathbf{r}) = V_{ext}(\mathbf{r}) + V_{Hartree}[n] + V_{xc}[n], \tag{12}$$

where $V_{Hartree}$ and V_{xc} are the Hartree potential and the exchange correlation potential, respectively. Figure 3 shows

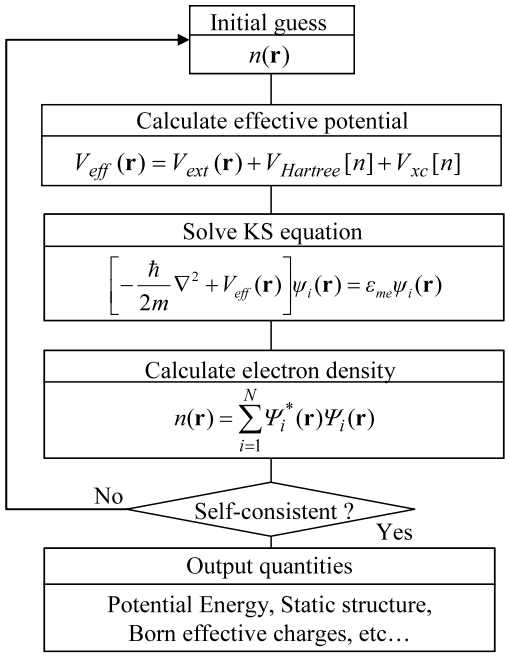


Fig. 3 Flow chart of solving the Kohn-Sham equation

the flow chart of solving the Kohn-Sham equation. First, the optional $n(\mathbf{r})$ is employed. The effective potential $V_{eff}(\mathbf{r})$ is estimated by employing the approximation such as the local density approximation (LDA) [76, 77] and the generalized gradient approximation (GGA) [78, 79] for the exchange correlation potential. The wave function $\Psi(\mathbf{r})$ is determined by using the obtained $V_{eff}(\mathbf{r})$. Then $\Psi(\mathbf{r})$ is introduced into Eq. (8) to calculate the $n(\mathbf{r})$. This process is kept calculating until self-consistently converging.

In the case of calculating the ion state of the solid such as a crystal, a unit cell structure is effectively adopted or regularly arrayed periodic crystal lattice in the infinity field. Under the periodic condition, the wave function, which transfers with the real lattice vector $\mathbf{R}$, is described as,

$$\Psi(\mathbf{r}) = \Psi(\mathbf{r} + \mathbf{R}). \tag{13}$$

The wave function is described by using the wave vector $\mathbf{k}$.

$$\Psi_j(\mathbf{r} + \mathbf{R}) = e^{i\mathbf{k}\cdot\mathbf{R}}\Psi_j(\mathbf{r}). \tag{14}$$

It is called as the Bloch theorem. Generally, Eq. (2.24) is described in the following equation by employing the function $u_j^{\mathbf{k}}(\mathbf{r})$ with the translation symmetry.

$$\Psi_j(\mathbf{r} + \mathbf{R}) = e^{i\mathbf{k}\cdot\mathbf{r}}u_j^{\mathbf{k}}(\mathbf{r}). \tag{15}$$

Applying the Bloch theorem [80] to the DFT, the potential energy of the unit crystal structure can be calculated under the condition of the periodic boundary.

2.3 Crystal Growth Design

Considering an epitaxial growth of the crystal on a substrate, the lattice constants including a, b, c, θ_{ab}, θ_{bc} and θ_{ca} are changed because of the lattice mismatch with the substrate. These crystal structure changes can be determined by considering six components of mechanical strain in crystallographic coordinate system such as ε_a, ε_b, ε_c, γ_{ab}, γ_{bc} and γ_{ca}. In a general analysis procedure, the lattice mismatch in the specific direction was calculated and the crystal growth potential was derived. However, the epitaxially grown thin film crystal is in a multi-axial state. Therefore, the numerical results of the crystal energy of thin films are not corrected when considering only uni-axis strain.

In this study, the total energy of a crystal thin film with multi-axial crystal strain states is calculated by using the first-principles calculation, and is applied to the case of the epitaxial growth process. An ultra-soft pseudo-potentials method is employed in the DFT with the condition of the LDA for exchange and correlation terms. Total energies of the thin film crystal as the function of six components of crystal strain are calculated to find a minimum value. Total energies are calculated discretely and a continuous function approximation is introduced. A sampling area is selected by considering the symmetry between a and b axes in a tetragonal crystal structure. Sampling points are generated by using a Latin hypercube sampling (LHS) method [81], which is the efficient tool to get nonoverlap sampling points. The following global function model is generated by using a kriging polynomial hybrid approximation (KPHA) method [82].

$$E = A_h\varepsilon_h^2 + B_{ij}\varepsilon_i\varepsilon_j + C_l\varepsilon_l + E_{T0}$$
$$(h, i, j = a, b, c, ab, bc, ca), \tag{16}$$

where E_{T0} is the total energy of the stable crystal, ε_h, ε_i and ε_j epitaxial strains and A_h, B_{ij} and C_l coefficients generated by KPHA method.

A gradient of total energy at each sampling point is calculated to generate an approximate quadratic function. The minimum point of a total energy can be found by using this function.

In the process crystallographic simulation, it is assumed that several crystal unit cells of crystal clusters, which have certain conformations, can grow on a substrate as shown Fig. 4. The left-hand side diagram in Fig. 9 shows an example of conformation in cases of [001], [100], [110] and [101] orientations, and the right-hand side shows [111] orientation. O, A and B are points of substrate atoms corresponding to thin films ones within the allowable range of distance. l_{OA} and l_{OB} indicate distances of A and B from O, respectively. θ_{AOB} indicates the angle between lines OA and OB.

Table 2 summarizes the relationship between the lattice constants of the thin film and l_{OA} and l_{OB} according to crystal orientations. Additionally, Table 2 shows crystal strains, which can be determined in the corresponded crystal orientations. However, particular crystal strains, such as ε_i^* and γ_{ij}^*, cannot be determined by employing the lattice constants of the thin film and the geometric constants of the substrate.

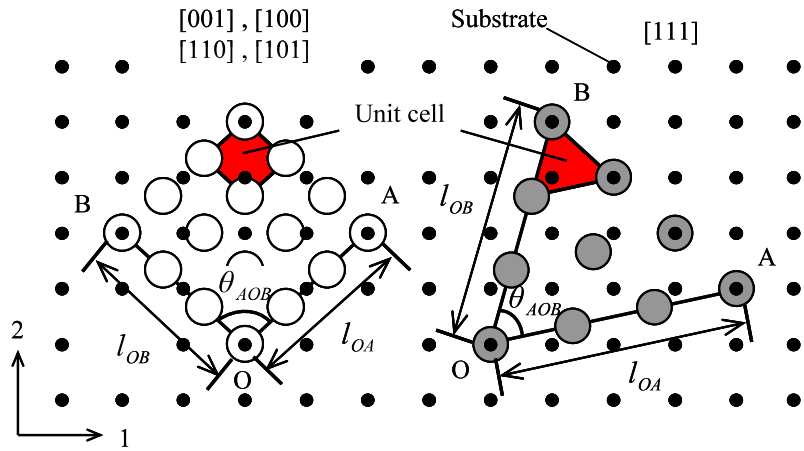


Fig. 4 Schematic of crystal conformations on a substrate

Table 2 Relationship of lattice constants and epitaxial strain with crystal orientations

	[001]	[100]	[110]	[101]	[111]
l_{OA}	a	c	c	$\sqrt{a^2+c^2}$	$\sqrt{a^2+c^2}$
l_{OB}	b	b	$\sqrt{a^2+b^2}$	b	$\sqrt{b^2+c^2}$
Epitaxial strain					
ε_a	$\frac{l_{OA}}{ka}-1$	ε_a^*	ε_a^*	ε_a^*	$\frac{\sqrt{(l_{OA}/k)^2-c^2(1+\varepsilon_c)^2}}{a}-1$
ε_b	$\frac{l_{OB}}{kb}-1$	$\frac{l_{OB}}{kb}-1$	$\frac{\sqrt{(l_{OB}/k)^2-a^2(1+\varepsilon_a^*)^2}}{b}-1$	$\frac{l_{OB}}{kb}-1$	$\frac{\sqrt{(l_{OB}/k)^2-c^2(1+\varepsilon_c)^2}}{b}-1$
ε_c	ε_c^*	$\frac{l_{OA}}{kc}-1$	$\frac{l_{OA}}{kc}-1$	$\frac{\sqrt{(l_{OA}/k)^2-a^2(1+\varepsilon_a^*)^2}}{c}-1$	$\frac{1}{c}\sqrt{(\frac{l_{OA}}{k}-\sqrt{a^2+c^2})(\frac{l_{OB}}{k}-\sqrt{b^2+c^2})\cos\theta_{AOB}}-1$
γ_{ab}	$\theta_{AOB}-\theta_{ab}$	γ_{ab}^*	γ_{ab}^*	$\theta_{AOB}-\theta_{ab}$	γ_{ab}^*
γ_{bc}	γ_{bc}^*	γ_{bc}^*	$\theta_{AOB}-\theta_{bc}$	γ_{bc}^*	$\frac{b}{c}\gamma_{ab}^*$
γ_{ca}	γ_{ca}^*	$\theta_{AOB}-\theta_{ca}$	$\theta_{AOB}-\theta_{ca}$	$\frac{a}{c}\gamma_{ab}$	$\frac{a}{c}\gamma_{ab}^*$

ε_i^* and γ_{ij}^* can be given from first-principles calculation to the minimize total energy

In this numerical analysis scheme, their unknown components are determined by employing the condition of minimum total energy of the crystal unit cell.

Figure 5 shows the flowchart of the crystal growth prediction algorithm. First, lattice constants of the thin film and the substrate are inputted. The following procedure is demonstrated. Substrate coordinates of A and B points, which are indicated as (m_A, n_A) and (m_B, n_B), are updated according to the numerical result under the condition of fixing O point in order to generate candidate crystal clusters with assumed conformations and orientations. The search range of the crystal cluster is settled as $0 < m_A, m_B < m$ and $0 < n_A, n_B < n$ by considering the grain size of the piezoelectric thin film crystal. $\vec{e}_1$ and $\vec{e}_2$ as shown in Fig. 5 are unit vectors of the substrate coordinate system. Lattice constants of the crystal cluster are compared with geometrical parameters of the substrate, and candidate crystal clusters, which have extreme lattice mismatches, are eliminated. Crystal strains caused by the epitaxial growth are calculated for every candidates of the grown crystal cluster as shown in Table 2. The total energy of grown crystal cluster is estimated by using the total energy as a function of crystal strains. Total energies of candidate crystal clusters are compared with one of the free-strained boundary condition, in order to calculate total energy increments of candidate crystal clusters.

The fraction of crystal cluster grown on the substrate is calculated by a canonical distribution [83].

$$p_i = \frac{\exp[-\Delta E_i/k_B T]}{\sum_n \exp[-\Delta E_n/k_B T]}, \tag{17}$$

where ΔE is the total energy increment of the grown cluster, k_B the Boltzmann constant and T the temperature.

2.4 Crystallographic Homogenization Method

The Crystallographic homogenization method scales up micro heterogeneous structure, such as polycrystalline aggregation, to macro homogeneous structure, such as continuum body. The micro heterogeneous structure has the area Y and microscopic polycrystalline coordinate y, and the macro homogeneous structure has the area X and macroscopic sample coordinate x. Here, it relates to two scales by using the scale ratio λ_h.

$$\lambda_h = \frac{x}{y}, \tag{18}$$

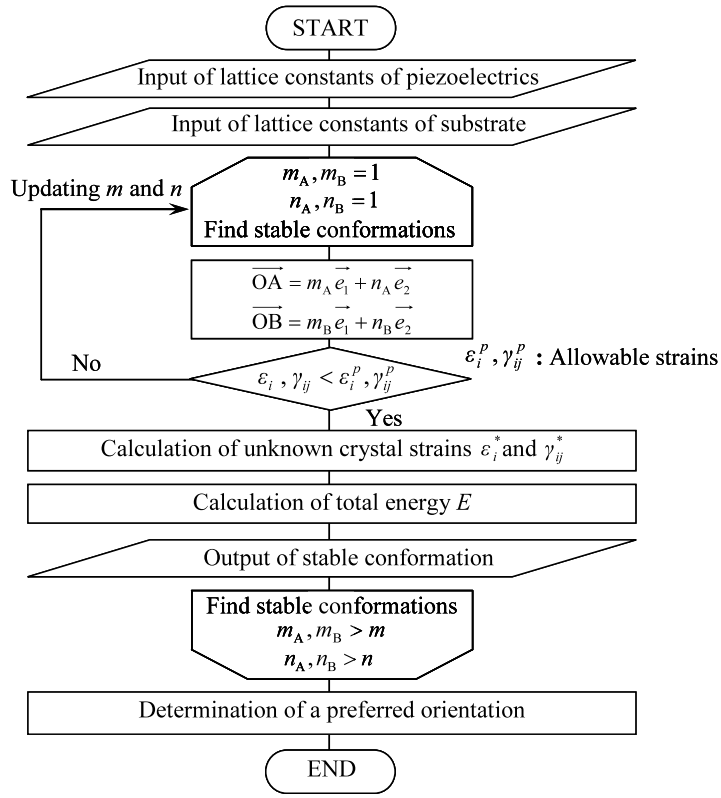


Fig. 5 Flowchart of the process crystallographic simulation

where λ_h is an extremely small value. Both coordinates of the micro polycrystalline and the macro continuum structures can be selected independently based on the Eq. (18).

For the macro continuum structure, the relationship between the strain $\varepsilon_{ij}^{\lambda_h}$ and the displacement $u_i^{\lambda_h}$ is assumed as follows;

$$\varepsilon_{ij}^{\lambda_h} = \frac{1}{2}\left(\frac{\partial u_i^{\lambda_h}}{\partial x_j} + \frac{\partial u_j^{\lambda_h}}{\partial x_i}\right). \tag{19}$$

Here coupling variables are affixed to the superscript λ_h, because the behavior of the piezo-elastic materials is affected by the polycrystalline structure and λ_h. The mathematical relationship between the electric field $E_i^{\lambda_h}$ and the electrostatic potential ϕ^{λ_h} is written as,

$$E_i^{\lambda_h} = -\frac{\partial \phi^{\lambda_h}}{\partial x_i}. \tag{20}$$

The balanced equations of the static strain $\sigma_{ij}^{\lambda_h}$ and the electro displacement $D_i^{\lambda_h}$, of which body forces can be neglected, are given by

$$\frac{\partial \sigma_{ij}^{\lambda_h}}{\partial x_i} = 0, \tag{21}$$

$$\frac{\partial D_i^{\lambda_h}}{\partial x_i} = 0. \tag{22}$$

For the plane $\Gamma_d^{\lambda_h}$, which is applied to surface force t_i on the surface of the macro continuum structure Γ^{λ_h}, and the plane $\Gamma_e^{\lambda_h}$ with the surface charge ρ are written as,

$$t_i = \sigma_{ij}^{\lambda_h} n_j \quad \text{on } \Gamma_d^{\lambda_h}, \tag{23}$$

$$\rho = D_i^{\lambda} n_i \quad \text{on } \Gamma_d^{\lambda_h}. \tag{24}$$

For the linear combination of the strain and the electric field, the linear piezo-elastic constitutive equation is described as,

$$\sigma_{ij}^{\lambda_h} = C_{ijkl}^{E\lambda_h} \varepsilon_{kl}^{\lambda_h} - e_{kij}^{\lambda_h} E_k^{\lambda_h}, \tag{25}$$

$$D_i^{\lambda_h} = e_{ikl}^{\lambda_h} \varepsilon_{kl}^{\lambda_h} + \epsilon_{ik}^{S\lambda_h} E_k^{\lambda_h}. \tag{26}$$

Equations (19)–(24) are introduced into the equation of the virtual work, and are applied to the Gauss's divergence theorem. Introducing Eqs. (25) and (26) into these equations, the dominant equations are written as,

$$\int_{\Omega^{\lambda_h}} \left(C_{ijkl}^{E\lambda_h} \varepsilon_{kl}^{\lambda_h} - e_{kij}^{\lambda_h} E_k^{\lambda_h}\right) \frac{\partial \delta u_i^{\lambda_h}}{\partial x_j} d\Omega + \int_{\Omega^{\lambda}} \left(e_{ikl}^{\lambda_h} \varepsilon_{kl}^{\lambda_h} + \epsilon_{ik}^{S\lambda_h} E_k^{\lambda_h}\right) \frac{\partial \delta \phi^{\lambda_h}}{\partial x_i} d\Omega = \int_{\Gamma_d^{\lambda}} t_i \delta u_i^{\lambda_h} d\Gamma + \int_{\Gamma_e^{\lambda}} \rho \delta \phi^{\lambda_h} d\Gamma. \tag{27}$$

The asymptotic expansion of the displacement and the electrostatic potential of the macro continuum structure with periodical micro polycrystalline structure are

$$u_i^{\lambda_h} = u_i^0(x) + \lambda_h u_i^1(x, y) + \cdots, \tag{28}$$

$$\phi^{\lambda_h} = \phi^0(x) + \lambda_h \phi^1(x, y) + \cdots. \tag{29}$$

where $u_i^0(x)$ and $\phi^0(x)$ are the displacement vector and the electrostatic potential of the macro continuum structure, respectively. $u_i^1(x, y)$ and $\phi^1(x, y)$ are the displacement vector and the electrostatic potential in the micro polycrystalline structure, respectively. In the case of considering the first and second term in Eqs. (28) and (29), the strain tensor and the electric field tensor are

$$\varepsilon_{ij}^{\lambda_h} = \frac{1}{2}\left(\frac{\partial u_i^0(x)}{\partial x_j} + \frac{\partial u_j^0(x)}{\partial x_i}\right) + \frac{1}{2}\left(\frac{\partial u_i^1(x, y)}{\partial y_j} + \frac{\partial u_j^1(x, y)}{\partial y_i}\right), \tag{30}$$

$$E_i^{\lambda_h} = -\frac{\partial \phi^0(x)}{\partial x_i} - \frac{\partial \phi^1(x, y)}{\partial y_i}. \tag{31}$$

It is assumed that the microscopic displacement and the electrostatic potential can be written as the separation of variables:

$$u_i^1(\boldsymbol{x}, \boldsymbol{y}) = \chi_i^{kl}(\boldsymbol{x}, \boldsymbol{y})\frac{\partial u_k^0(\boldsymbol{x})}{\partial x_l} + \Phi_i^m(\boldsymbol{x}, \boldsymbol{y})\frac{\partial \phi^0(\boldsymbol{x})}{\partial x_m}, \tag{32}$$

$$\phi^1(\boldsymbol{x}, \boldsymbol{y}) = \varphi^{ij}(\boldsymbol{x}, \boldsymbol{y})\frac{\partial u_i^0(\boldsymbol{x})}{\partial x_j} + R^k(\boldsymbol{x}, \boldsymbol{y})\frac{\partial \phi^0(\boldsymbol{x})}{\partial x_k}, \tag{33}$$

where $\chi_i^{kl}(x, y)$ is the characteristic displacement of a unit cell, $R^k(x, y)$ the characteristic electrical potential of a unit cell and $\phi^{ij}(x, y)$ and $\Phi_i^m(x, y)$ the characteristic coupling functions of a unit cell. Applying the averaging theorem to Eq. (27), Eqs. (28)–(33) are described as,

$$\int_Y \left(C_{ijkl}^E \frac{\partial \chi_k^{mn}(x, y)}{\partial y_l} + e_{kij}\frac{\partial \varphi^{mn}(x, y)}{\partial y_k}\right)\frac{\partial \delta u_i^1(x, y)}{\partial y_j} dY = -\int_Y C_{ijmn}^E \frac{\partial \delta u_i^1(x, y)}{\partial y_j} dY, \tag{34}$$

$$\int_Y \left(e_{ikl}\frac{\partial \chi_k^{mn}(x, y)}{\partial y_l} - \epsilon_{ik}^S \frac{\partial \varphi^{mn}(x, y)}{\partial y_k}\right)\frac{\partial \delta \phi^1(x, y)}{\partial y_i} dY = -\int_Y e_{imn}\frac{\partial \delta \phi^1(x, y)}{\partial y_i} dY, \tag{35}$$

$$\int_Y \left(C_{ijkl}^E \frac{\partial \Phi_k^p(x, y)}{\partial y_l} + e_{kij}\frac{\partial R^P(x, y)}{\partial y_k}\right)\frac{\partial \delta u_i^1(x, y)}{\partial y_j} dY = -\int_Y e_{pij}\frac{\partial \delta u_i^1(x, y)}{\partial y_j} dY, \tag{36}$$

$$\int_Y \left(e_{ikl}\frac{\partial \Phi_k^p(x, y)}{\partial y_l} - \epsilon_{ik}^S \frac{\partial R^P(x, y)}{\partial y_k}\right)\frac{\partial \delta \phi^1(x, y)}{\partial y_i} dY = \int_Y \epsilon_{ip}^S \frac{\partial \delta \phi^1(x, y)}{\partial y_i} dY. \tag{37}$$

Equations (34)–(37) have the solution under the condition of the periodic boundary. The homogenized elastic stiffness tensor, piezoelectric stress constant tensor and dielectric constant tensor are described by the following characteristic function tensor.

$$C_{ijmn}^{EH} = \frac{1}{|Y|}\int_Y \left(C_{ijmn}^E + C_{ijkl}^E \frac{\partial \chi_k^{mn}(x, y)}{\partial y_l} + e_{kij}\frac{\partial \varphi^{mn}(x, y)}{\partial y_k}\right) dY, \tag{38}$$

$$e_{pij}^H = \frac{1}{|Y|}\int_Y \left(e_{pij} + e_{kij}\frac{\partial R^P(x, y)}{\partial y_k} + C_{ijkl}^E \frac{\partial \Phi_k^p(x, y)}{\partial y_l}\right) dY = \frac{1}{|Y|}\int_Y \left(e_{pij} + e_{pkl}\frac{\partial \chi_k^{ij}(x, y)}{\partial y_l} - \epsilon_{pk}^S \frac{\partial \varphi^{ij}(x, y)}{\partial y_k}\right) dY, \tag{39}$$

$$\epsilon_{ip}^{SH} = \frac{1}{|Y|}\int_Y \left(\epsilon_{ip}^S + \epsilon_{ik}^S \frac{\partial R^P(x, y)}{\partial y_k} - e_{ikl}\frac{\partial \Phi_k^p(x, y)}{\partial y_l}\right) dY, \tag{40}$$

where superscript H means the homogenized value. The macroscopic balanced equations are given by utilizing Eqs. (38)–(40).

$$\int_\Omega \left(C_{ijkl}^{EH}\frac{\partial u_k^0(x)}{\partial x_l} + e_{kij}^H \frac{\partial \phi^0(x)}{\partial x_k}\right)\frac{\partial \delta u_i^0(x)}{\partial x_j} d\Omega = \int_{\Gamma_d} t_i \delta u_i^0(x) d\Gamma, \tag{41}$$

$$\int_\Omega \left(e_{ikl}^H \frac{\partial u_k^0(x)}{\partial x_l} - \epsilon_{ik}^{SH}\frac{\partial \phi^0(x)}{\partial x_k}\right)\frac{\partial \delta \phi^0(x)}{\partial x_i} d\Omega = \int_{\Gamma_e} \rho \delta \phi^0(x) d\Gamma. \tag{42}$$

Additionally, introducing Eq. (32) and Eq. (33) into Eq. (28) and Eq. (29), the displacement and the electrostatic potential of overall structure are described as,

$$u_i^\lambda = u_i^0(x) + \lambda_h\left(\chi_i^{mn}\frac{\partial u_m^0(x)}{\partial x_n} + \Phi_i^p \frac{\partial \phi^0(x)}{\partial x_p}\right), \tag{43}$$

$$\phi^\lambda = \phi^0(x) + \lambda_h\left(\varphi^{mn}\frac{\partial u_m^0(x)}{\partial x_n} + R^p \frac{\partial \phi^0(x)}{\partial x_p}\right). \tag{44}$$

In this study, the scale ratio is employed to the zero based on the assumption that the macro continuum structure is

infinitely and periodically at one point in the macro continuum structure. Therefore, Assuming the micro polycrystalline structure is extremely small for the macro continuum structure and the assumption is satisfied, the macroscopic displacement and the electrostatic potential are written as,

$$u_i^{\text{macro}}(x) = u_i^0(x), \tag{45}$$

$$\phi^{\text{macro}}(x) = \phi^0(x). \tag{46}$$

The macroscopic displacement and electrostatic potential are evaluated by following equations based on a uniform displacement and the electrostatic potential of the macro continuum structure.

$$u_i^{\text{micro}}(x, y) = y_j \frac{\partial u_i^0(x)}{\partial x_j} + \chi_i^{mn}(x, y) \frac{\partial u_m^0(x)}{\partial x_n} + \Phi_i^p(x, y) \frac{\partial \phi^0(x)}{\partial x_p}, \tag{47}$$

$$\phi^{\text{micro}}(x, y) = y_j \frac{\partial \phi^0(x)}{\partial x_j} + \varphi^{mn}(x, y) \frac{\partial u_m^0(x)}{\partial x_n} + R^p(x, y) \frac{\partial \phi^0(x)}{\partial x_p}. \tag{48}$$

The microscopic strain tensor, electric field vector and stress tensor are described by using Eqs. (47)–(48).

$$\varepsilon_{ij}^{\text{micro}}(x, y) = \frac{\partial u_i^{\text{micro}}(x, y)}{\partial y_j}, \tag{49}$$

$$E_i^{\text{micro}}(x, y) = -\frac{\partial \phi^{\text{micro}}(x, y)}{\partial y_i}, \tag{50}$$

$$\sigma_{ij}^{\text{micro}}(x, y) = C_{ijkl}^E(x, y) \varepsilon_{kl}^{\text{micro}}(x, y) - e_{kij}(x, y) E_k^{\text{micro}}(x, y). \tag{51}$$

The conventional isoparametric finite element is employed for discretized modeling of micro polycrystalline and macro continuum structures to decrease the degree of freedom. For three-dimensional hexahedral isoparametric element, such as the linear element with eight nodes, the two-order element with twenty nodes and the three-order element with thirty two nodes, have been proposed. In this study, the simplest linear element with eight nodes is employed.

The physical and mathematical modeling of the three-scale structure and the numerical analysis scheme of three-scale analysis was newly developed. The conventional two-scale finite element analysis is based on the crystallographic homogenization method. In this conventional analysis, the virtually determined or experimentally measured orientations are employed for the micro crystalline structure to characterize the macro homogenized piezoelectric properties. However, this conventional analysis can not characterize a new piezoelectric thin film because of unknown crystal structure and material properties. A newly proposed three-scale structure analysis can scale up and characterize the crystal structure to the micro polycrystalline and macro continuum structures. First, the stable structure and properties of the new piezoelectric crystal were evaluated by using the first-principles DFT. Second, the crystal growth process of the new piezoelectric thin film was analyzed by using the process crystallographic simulation. The preferred orientation and their fractions of the micro polycrystalline structure were predicted by this simulation. Finally, the homogenized piezoelectric properties of the macro continuum structure were characterized by using the crystallographic homogenization theory. Comparing the provability of crystal growth and the homogenized piezoelectric properties of the new piezoelectric thin film on several substrates, the best substrate was found by using the three-scale structure analysis. It was confirmed that the three-scale structure analysis could design not only existing thin films but also new piezoelectric thin films.

2.5 Solving Methodology for Coupling Problem

In this section, we present a parallel computing procedure for the multi-scale piezoelectric analysis based on the crystallographic homogenization method. A large-scale FE equation is necessary to solve fine FE mesh, such as the EBSD measured realistic voxel micro-FE model in the crystallographic homogenization method [84] or the thin film structure of macro-FE model. The iterative solver based on the Kryrov subspace method, such as the conjugate gradient (CG) method, is suitable to large-scale FE analysis with parallel computing technique. In the FE equation for the piezoelectric analysis, however, the CG method cannot apply to the piezoelectric analysis because the coefficient matrix of coupling FE equation,

$$\begin{bmatrix} \mathbf{K}_{uu} & \mathbf{K}_{u\phi} \\ \mathbf{K}_{\phi u} & -\mathbf{K}_{\phi\phi} \end{bmatrix} \begin{Bmatrix} \mathbf{u} \\ \phi \end{Bmatrix} = \begin{Bmatrix} \mathbf{t} \\ \mathbf{q} \end{Bmatrix}, \tag{52}$$

is not positive definite due to the negative value of the sub-matrix $\lfloor \mathbf{K}_{\phi\phi} \rfloor$, in which the system matrix has negative eigenvalue. The system matrix is symmetric because of $[\mathbf{K}_{u\phi}] = [\mathbf{K}_{\phi u}]^T$. But the coupling matrix is strongly ill-condition that the condition number may be over 10^{20} because huge different numerical order 10^{-13} to 10^{11} in the coefficient matrix may appear in actual analysis between mechanical and electrical fields.

Until now, some iterative methodologies for piezoelectric coupling analysis were developed as summarized in Table 3. In the vector potential method [85, 86], the h-type piezoelectric elastic constitutive equation,

$$\sigma_{ij} = C_{ijkl}^D \varepsilon_{kl} + h_{mij} D_m, \tag{53}$$

$$E_n = -h_{nij} \varepsilon_{ij} + \beta_{nm}^S D_m, \tag{54}$$

Table 3 Comparison of solving methodologies for piezoelectric coupling problem

problem	Methods	Characteristics
Non-coupling	LU decomposition Gaussian elimination CG method	• Positive definite • Symmetric matrix
Coupling	Vector potential method (By using vector potential to electric filed)	• Semi-positive definite • Symmetric matrix • Gage condition by using the edge element
	Preconditioning method (Changing diagonal terms to positive value of electric field and apply the precondition to the matrix)	• Positive definite • Asymmetric matrix • Preconditioning
	Iterative partitioned method (Splitting unknown vectors into mechanical and electric fields based on the block Jaccobi method or the block Gauss-Seidel method)	• Positive definite • Symmetric matrix • Convergence depends on piezoelectric material properties

is used for analysis instead of e-type constitutive equations (1) and (2), where h_{nij} is the piezoelectric stress constant tensor. The system matrix of FE equation by using the h-type constitutive equation is semi-positive definite. Since the unknown for electric filed is the electric vector potential $\boldsymbol{\psi}$, the gage condition by using the edge element is used in the 3-D problem.

On the other hand, the FE equation can be rewritten as

$$\begin{bmatrix} \mathbf{K}_{uu} & \mathbf{K}_{u\phi} \\ -\mathbf{K}_{\phi u} & \mathbf{K}_{\phi\phi} \end{bmatrix} \begin{Bmatrix} \mathbf{u} \\ \phi \end{Bmatrix} = \begin{Bmatrix} \mathbf{t} \\ -\mathbf{q} \end{Bmatrix}, \tag{55}$$

to maintain positive definite of the coefficient matrix, but, which is asymmetric. Therefore, preconditioned Kryrov subspace solver for asymmetric matrix, such as Bi-CG method, is applied to the FE equation [87].

In this study, a parallel computing technique for piezoelectric FE analysis is newly developed based on the iterative partitioned coupling method with the parallel conjugate gradient (CG) solver [88]. In the iterative partitioned method, the system equation is separated into mechanical and electrical fields based on the block Jaccobi method or the block Gauss-Seidel method. Since both coefficient matrices of the separated equation are positive definite and the condition number is dramatically decreased, the CG solver can be individually applied to the separated equations without the preconditioning.

We assume that the piezoelectric material may have averaged properties in a macro continuum body. The characteristic function vectors $\{\boldsymbol{\chi}^{mn}\}$, $\{\varphi^{mn}\}$, $\{\boldsymbol{\Phi}^{p}\}$ and $\{\mathbf{R}^{p}\}$ are obtained by solving the microscopic FE equations for the RVE as follows:

$$\begin{bmatrix} \mathbf{K}_{uu} & \mathbf{K}_{u\phi} \\ \mathbf{K}_{\phi u} & -\mathbf{K}_{\phi\phi} \end{bmatrix} \begin{Bmatrix} \boldsymbol{\chi}^{mn} \\ \varphi^{mn} \end{Bmatrix} = \begin{Bmatrix} \mathbf{t}^{mn} \\ \mathbf{q}^{mn} \end{Bmatrix}, \tag{56}$$

$$\begin{bmatrix} \mathbf{K}_{uu} & \mathbf{K}_{u\phi} \\ \mathbf{K}_{\phi u} & -\mathbf{K}_{\phi\phi} \end{bmatrix} \begin{Bmatrix} \boldsymbol{\Phi}^{p} \\ \mathbf{R}^{p} \end{Bmatrix} = \begin{Bmatrix} \mathbf{t}^{p} \\ \mathbf{q}^{p} \end{Bmatrix}, \tag{57}$$

where the characteristic functions have 9 components such as $(m, n) = (1,1), (2,2), (3,3), (2,3), (3,1), (1,2)$, and $p = 1, 2, 3$. The right-hand side vectors $\{\mathbf{t}^{mn}\}$, $\{\mathbf{q}^{mn}\}$, $\{\mathbf{t}^{p}\}$ and $\{\mathbf{q}^{p}\}$ are the constant values caused by the microscopic inhomogeneous structure. The size of the linear equation depends on the number of nodal points in the RVE FE model, and the degrees of freedom may be a large number for the SEM-EBSD measured model.

In order to apply the CG method to solve the system equations, they are rewritten to a partitioned form based on the Block Gauss-Seidel (BGS) method as follow:

$$\begin{bmatrix} \mathbf{K}_{uu} & \mathbf{0} \\ -\mathbf{K}_{\phi u} & \mathbf{K}_{\phi\phi} \end{bmatrix} \begin{Bmatrix} \mathbf{u} \\ \phi \end{Bmatrix}^{(k+1)} = \begin{Bmatrix} \mathbf{t} \\ -\mathbf{q} \end{Bmatrix} - \begin{bmatrix} \mathbf{0} & \mathbf{K}_{u\phi} \\ \mathbf{0} & \mathbf{0} \end{bmatrix} \begin{Bmatrix} \mathbf{u} \\ \phi \end{Bmatrix}^{(k)}, \tag{58}$$

where unknown vector $\{\mathbf{u}\}$ corresponds to $\{\boldsymbol{\chi}^{mn}\}$ and $\{\boldsymbol{\Phi}^{p}\}$, and $\{\phi\}$ corresponds to $\{\varphi^{mn}\}$ and $\{\mathbf{R}^{p}\}$, respectively. According the coupling term $\lfloor \mathbf{K}_{u\phi} \rfloor$ is moved to right hand side, displacement $\{\mathbf{u}\}$ and electrostatic potential $\{\phi\}$ vectors are partitioned to maintain positive definite of both the coefficient matrices as follows:

$$[\mathbf{K}_{uu}]\{\mathbf{u}\}^{(k+1)} = \{\mathbf{t}\} - [\mathbf{K}_{u\phi}]\{\phi\}^{(k)}, \tag{59}$$

$$[\mathbf{K}_{\phi\phi}]\{\phi\}^{(k+1)} = -\{\mathbf{q}\} + [\mathbf{K}_{\phi u}]\{\mathbf{u}\}^{(k+1)}. \tag{60}$$

Since both coefficient matrices $[\mathbf{K}_{uu}]$ and $\lfloor \mathbf{K}_{\phi\phi} \rfloor$ are symmetric and positive definite, the CG solver can apply to the linear equations, respectively. If the sub-matrix $\lfloor -\mathbf{K}_{\phi u} \rfloor$ is also moved to the right hand side, it is the block Jaccobi method. The number of iterations in the block Jaccobi method is larger than the BGS method.

The parallel CG solver [89], which is parallelized on the inner product of the coefficient matrix and the descent direction vector based on row block distribution of the coef-

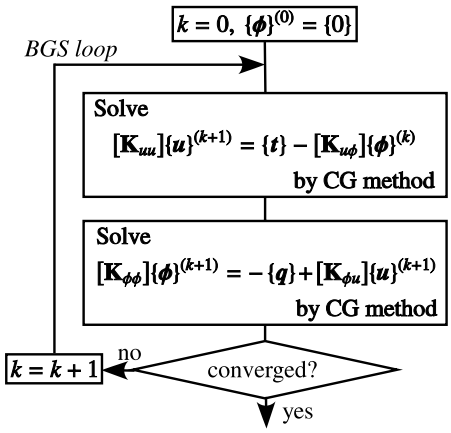


Fig. 6 Flowchart of iterative partitioned coupling procedure based on the block Gauss-Seidel (BGS) method

ficient matrix, is applied to each partitioned equation until unknown vectors $\{\mathbf{u}\}$ and $\{\phi\}$ are converged with mutually substituting the coupling terms based on the BGS method as shown in Fig. 6. A nested iteration procedure of the BGS method including the CG iteration is constructed in this scheme.

In this study, a hierarchical process distribution technique is introduced to reduce amount of data for communication. A cluster of PCs is logically divided into 9 processor groups, and allocated to the micro FE equations using the partitioning form. In each processor group, the parallel CG solver based on the block partitioning is applied to the linear equation. The parallel analysis code is implemented by using the Fortran 90 language and the message passing interface (MPI) library MPICH 1.2 [90].

3 Process Crystallographic Design

In this chapter, the application of three-scale structure design and analysis scheme was presented to discover and generate a new biocompatible piezoelectric thin film. The three-scale structure design and analysis begins from the biocompatible element selection [91] and it's followed by crystal structure design [92] and crystal growth design [93]. Finally it leads to material property evaluation of new biocompatible piezoelectric thin films [94].

3.1 Biocompatible Element Selection

After piezoelectric materials are dissolved and produce metal ions in body fluid, they may have the following influence on a living body. (a) Destruction of biological functions caused by bonging with proteins. (b) Genetic damage caused by reaction with DNA (deoxyribonucleic acid). (c) Biological lesion caused by substitution of bio-essential elements. In this study, substituting and competing elements (Pb, Tl, Cd, Sr, W) and catalytic toxicity elements (Hg, Cs, Ru, Rh, Pd, Os, Ir, Pt, Co), that are classified into type (c), are excluded from constituent elements. On the other hand, bio-essential elements (Ca, Cr, Cu, Fe, Ge, Mg, Mn, Mo, Na, Ni, Sn, V, Zn) and already-known in-vivo elements (Si, Ca, Ta, Ti, Zr) are added to constituent elements with no evaluation of interaction energy. And then the biocompatibility is estimated from the interaction energy for metal elements (Au, Ba, Be, Bi, In, Li, Rb, Re, Sc, Tc, Y) that may have the influence such as types (a) and (b).

Metal ions dissolved from piezoelectric materials are in danger of biological and genetic damage caused by reaction with in-vivo molecules such as proteins and DNA. The reaction between metal ions and in-vivo molecules can be estimated as a chemical reaction between a Lewis acid and a Lewis base. Consequently, in order to evaluate the reactivity of metal ions on proteins and DNA, we employ the HSAB principle which states that "hard acids prefer to coordinate to hard bases and soft acids to soft bases" [95, 96]. The hardness or softness (inverse of hardness) of species can be defined quantitatively by using the idea of polarizability. A less polarizable species is hard and a more easily polarized one is soft. As an indicator of reactivity, HSAB principle presents the interaction energy between acid and base which is defined by each electronegativity and hardness. In this study, the interaction energy between metal ions and important in-vivo molecules is evaluated quantitatively by HSAB principle, and little-reactionary ions are distinguished to find biocompatible elements.

The absolute electronegativity χ_S of a neutral species S is defined by the energy E_S and the number of electron N_S, and its negative is equal rigorously to the chemical potential of electrons μ_S as shown in Eq. (61).

$$\chi_S = -\left(\frac{\partial E_S}{\partial N_S}\right) = -\mu_S \tag{61}$$

Additionally, the absolute electronegativity χ_S can be also expressed by Eq. (2) if ionization potential I_S and electron affinity A_S are given for the species S.

$$\chi_S = \frac{1}{2}(I_S + A_S) = -\frac{1}{2}(\varepsilon_{\mathrm{HOMO}} + \varepsilon_{\mathrm{LUMO}}) \tag{62}$$

Here, ionization potential I_S and electron affinity A_S are approximated by energy of HOMO (Highest Occupied Molecular Orbital) $\varepsilon_{\mathrm{HOMO}}$ and energy of LUMO (Lowest Unoccupied Molecular Orbital) $\varepsilon_{\mathrm{LUMO}}$, respectively. The energies of HOMO and LUMO can be obtained by the first principles DFT calculations. On the other hand, the chemical hardness η_S of the species S is defined by Eq. (63), and it can be also obtained from the energies of HOMO and LUMO.

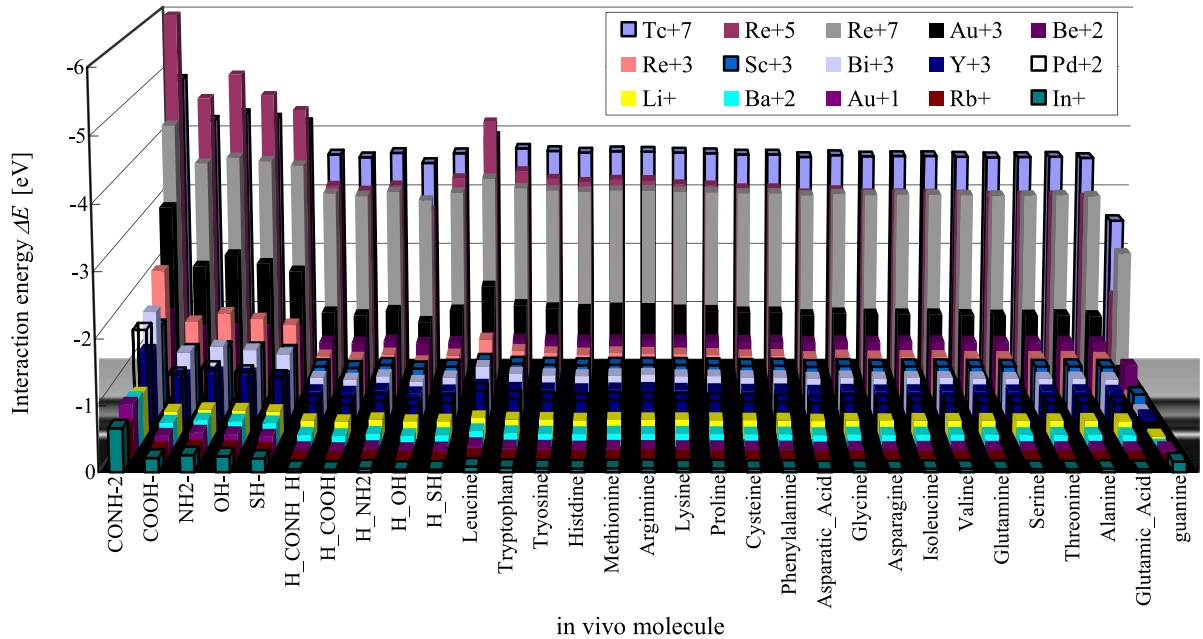


Fig. 7 Interaction energy between in vivo molecules and metal ions

Table 4 In vivo molecules used for bio-compatible evaluation

Anion of base	$CONH^{-2}$, $COOH^{-}$, NH^{2-}, OH^{-}, SH^{-}
Base terminated hydrogen	H-CONH-H, H-COOH, H-NH_2, H-OH, H-SH
Amino acid	Alanine, Argimine, Asparagine, Asparatic acid, Cysteine, Glutamic acid, Glutamine, Glycine, Histdine, Isoleucine, Leucine, Lysine, Methionine, Phenylalanine, Proline, Serine, Threonine, Tryosine, Tryptophan, Valine
DNA	Guanine

$$\eta_S = \frac{1}{2}\left(\frac{\partial^2 E_S}{\partial N_S^2}\right) = \frac{1}{2}(I_S - A_S) = -\frac{1}{2}(\varepsilon_{\text{HOMO}} - \varepsilon_{\text{LUMO}}) \tag{63}$$

Now, we consider a chemical reaction between acid A and base B. The reaction proceeds by electron transfer from base B to acid A at the overlap of HOMO and LUMO. The energy change ΔE associated to the electron transfer, that is the interaction energy, can be described as shown in Eq. (64).

$$\Delta E = (E_A - E_{A^0}) + (E_B - E_{B^0}) = -\frac{1}{2}(\chi_{A^0} - \chi_{B^0})\Delta N = -\frac{(\chi_{A^0} - \chi_{B^0})^2}{4(\eta_{A^0} + \eta_{B^0})} \tag{64}$$

Here, ΔN means the number of electron transfer. E_A, χ_A and η_A are energy, electronegativity and hardness for acid A, and the same notions are utilized for base B. Superscript 0 means the quantity of state before chemical reaction. When the interaction energy ΔE is more negative, A and B are more reactionary. In this study, metal ions and in-vivo molecules are set for acid A and base B, respectively. As the interaction energy ΔE is more negative for more toxic, bio-compatible elements are distinguished.

Typical in-vivo molecules, proteins and DNA were utilized for calculation of interaction energy to specify bio-compatible metal elements. There are a great many proteins, and they have large molecular number and complicated molecular structure. It is impossible to consider all kinds and overall structure of proteins into the first principles DFT calculations. Consequently, we focused on typical constituents of proteins and DNA basis, that are 20 amino acids, 5 anions of base, 5 bases terminated by hydrogen and easy-damaged guanine as shown in Table 4, and evaluated the bio-compatibility of metals from interaction energy with them.

To estimate the interaction energy between the above in-vivo molecules and metals, HOMO and LUMO energies of metal ions were calculated by using Gaussian 98 [97] with exception of toxicity and bio-essential elements. We utilized DFT (Becke & Lee, Yang, Parr: B3LYP/6-311++G(d, p)) for the 3rd or less periodic metal elements, and DFT (B3LYP/LANL2DZ) for the 4th or more periodic metal elements. With the same way, we utilized DFT (B3LYP/6-311++G(d, p)) for the in-vivo molecules, and calculated HOMO and LUMO energies. If compared with experimental values of ionization potential and electron affinity, calculational values of HOMO and LUMO energy

by using Gaussian 98 were verified to have average errors 3.09 % and 9.07 %, respectively.

Figure 7 shows the interaction energy between in-vivo molecules and metal ions obtained by DFT calculations. The interaction energy depends strongly on kind of metal ions, but it has little influence of in-vivo molecules. Therefore, the average value of interaction energy among in-vivo molecules can be useful to estimate overall biocompatibility of metal ions. On the other hand, Yamamoto A [98] employed IC50 value (the half maximal Inhibitory Concentration), which represents the concentration of an inhibitor that is required for 50 % inhibition of cells, and she has been reported the experimental results for cytotoxicity evaluation of 43 metals using murine fibroblasts and osteoblastic cells. To verify the validity of bio-compatibility obtained from DFT calculations, Fig. 8 shows the comparison with his cytotoxicity tests. The vertical axis means the average IC50 value obtained from cytotoxicity tests using two cells, and the cytotoxicity is strong at low IC50 value. The horizontal axis shows the interaction energy obtained from DFT calculations, and the chemical reactivity is high if the absolute value of interaction energy is large.

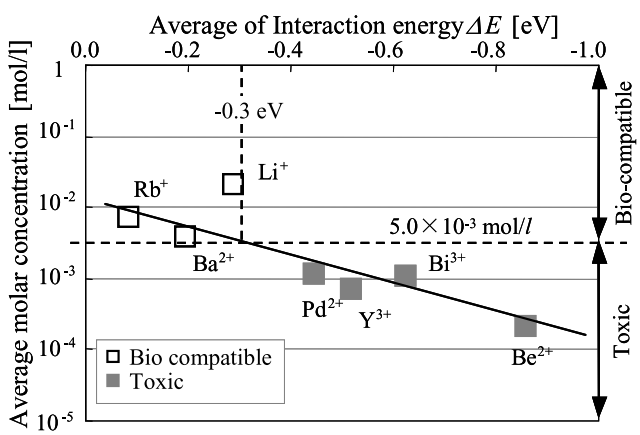


Fig. 8 Relation between experimental result of half maximal inhibitory concentration and calculational result of interaction energy

Figure 8 excludes toxicity and bio-essential metals from 43 metals, and summarizes 7 common metals between cytotoxicity tests and DFT calculations. As there is good correlation between both results, the interaction energy is an effective indicator enough to estimate bio-compatibility. The IC50 value of bio-materials applied to living body is more than 5.0×10^{-3} mol/l. If approximated linearly as shown in Fig. 6, the interaction energy corresponding to critical IC50 values is more than -0.3 eV. Consequently, metals were distinguished into bio-compatible elements if interaction energy is beyond -0.3 eV. As a result, 6 metals, Li^+, Ba^{2+}, K^+, Au^+, Rb^+, In^+ were distinguished into bio-compatible elements. We added bio-essential elements and living body-applied elements to bio-compatible elements, and employed them to constituents of bio-compatible piezoelectric material.

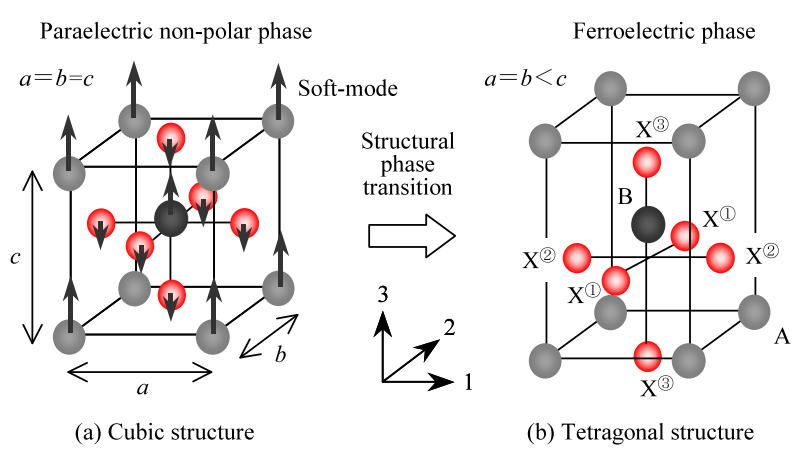
3.2 Crystal Structure Design

The perovskite-type compounds ABX_3 provide well-known examples of displacive phase transitions. They are in a paraelectric non-polar phase at high temperature and have a cubic crystal structure (lattice constant $a = c$) as shown in Fig. 9(a). The cubic crystal structure consists of A cations in the large eightfold coordinated site, B cations in the octahedrally coordinated site, and X anions at equipoint. The stability of cubic crystal structure can be estimated by an essential geometric condition, tolerance factor t. If ion radiuses of A, B and X are r_A, r_B, r_X, tolerance factor t can be described as

$$t = \frac{r_A + r_X}{\sqrt{2}(r_B + r_X)} \tag{65}$$

When tolerance factor t consists in the range from 0.75 to 1.10, the perovskite-type crystal structure has high stability. The cubic crystal structure often distorts to ferroelectric phase of lower symmetry at decreased temperature, that is a tetragonal crystal structure ($a > c$) with spontaneous polarization and strain as shown in Fig. 9(b). These ferroelectric

Fig. 9 Structural phase transition in perovskite-type crystal structure



distortions are caused by a soft-mode of phonon in cubic crystal structure, and it brings to good piezoelectricity. The soft-mode can be distinguished from other phonon modes with negative eigenfrequency, and the stable crystal structure after phase transition depends strongly on the eigenvectors.

Based on the nature of perovskite-type crystal structure, new biocompatible piezoelectric materials were searched according to the flowchart in Fig. 10. Firstly, biocompatible elements were inputted to A and B cations while oxygen was set to X anion for the perovskite-type compounds. The combination of three elements was determined to satisfy the stable condition of the tolerance factor.

The stable cubic structure of perovskite oxides was then obtained to minimize the total energy through first principles DFT calculations. The band structure was analyzed for the stable cubic structure to exclude conducting substances and find insulating ones. Next, the phonon properties of the stable cubic structure was analyzed for insulations to determine the soft-mode that causes a phase transition from the paraelectric non-polar phase (cubic structure) to the ferroelectric phase (tetragonal structure). When the eigenfrequency of phonon is positive, it is considered that the cubic structure is the most stable phase and don't transit to other phase. On the other hand, in case that the eigenfrequency is negative, the cubic structure is guessed to be an unstable phase and transit to other phase corresponding to the soft-mode. Consequently if phonon eigenfrequency was negative, noncentrosymmetric crystal structures were searched under no symmetry condition by using the eigenvector components for the initial atomic coordinates. We focused on tetragonal structure, which brings to superior piezoelectricity, and evaluated their material properties.

The biocompatible elements (Ca, Cr, Cu, Fe, Ge, Mg, Mn, Mo, Na, Ni, Sn, V, Zn, Si, Ta, Ti, Zr Li, Ba, K, Au, Rb, In), which have been specified by HSAB principle, were inputted to A and B cations in the perovskite compound ABX_3. Oxygen (O) was applied to X anion because oxides are easy to fabricate. The values of tolerance factor were calculated on the base of Pauling ionic radius. As a result, 28 oxides have even-numbered electrons and satisfied electric equilibrium condition and geometric stable condition. Then the band structure of stable cubic structure was calculated in order to exclude conducting sub-

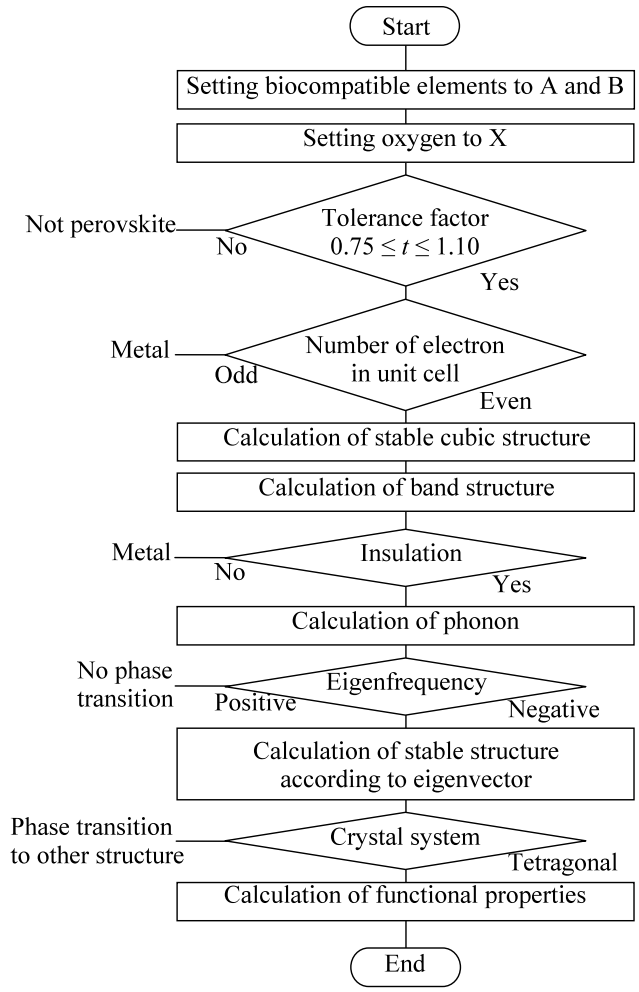


Fig. 10 The flowchart of searching new lead-free piezoelectric materials through the first-principles calculations

Table 5 The lattice constants and illustration of tetragonal structure for discovered lead-free perovskite oxides

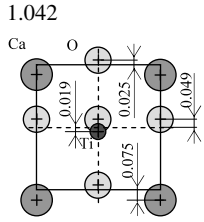
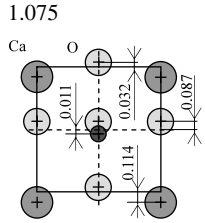
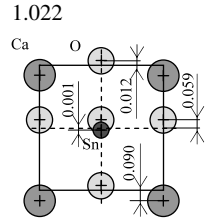
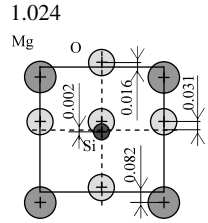
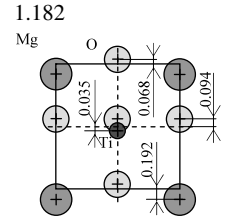
	$CaTiO_3$	$CaZrO_3$	$CaSnO_3$	$MgSiO_3$	$MgTiO_3$
Lattice constant [nm]	$a = 0.3778$ $c = 0.3939$	$a = 0.3978$ $c = 0.4275$	$a = 0.3949$ $c = 0.4036$	$a = 0.3448$ $c = 0.3534$	$a = 0.3673$ $c = 0.4343$
Aspect ratio c/a	1.042	1.075	1.022	1.024	1.182
Internal coordinates 3 ↑ ↳ 1					

Table 6 The comparison of spontaneous polarization and piezoelectric stress constant among discovered lead-free perovskite oxides

Material	Spontaneous polarization P^S [C/m^2]	Piezoelectric stress constant e_{333} [C/m^2]
$BaTiO_3$	0.220	6.99
$MgSiO_3$	0.464	5.61
$CaTiO_3$	0.630	5.24
$MgTiO_3$	1.33	2.47
$CaZrO_3$	0.723	2.89
$CaSnO_3$	0.437	2.23

stances. DFT calculations indicated 12 oxides, $CaGeO_3$, $MgGeO_3$, $CaSiO_3$, $MgSiO_3$, $BaTiO_3$, $CaTiO_3$, $MgTiO_3$, $BaZrO_3$, $CaZrO_3$, $BaSnO_3$, $CaSnO_3$, $MgSnO_3$, have band gap and are insulating substances.

The phonon properties of cubic structure at paraelectric non-polar phase were analyzed for the above 12 oxides. Calculations indicated 8 oxides have negative value of eigenfrequency and they become unstable by softening atomic vibration. Non-centrosymmetric crystal structures caused by soft mode were investigated. As a result, it was recognized that 6 oxides change to tetragonal structure. One of discovered 6 oxides, $BaTiO_3$ is well-known as a typical lead-free ferroelectric material. Table 5 summarizes lattice constants and illustration of stable tetragonal structure for the other 5 oxides, $CaTiO_3$, $CaZrO_3$, $CaSnO_3$, $MgSiO_3$, $MgTiO_3$. Table 6 shows the comparison of spontaneous polarization and piezoelectric stress constant e_{333} among the obtained lead-free perovskite oxides. Although the existing $BaTiO_3$ has the highest of 6 oxides, $MgSO_3$ and $CaTiO_3$ have higher piezoelectric constant than other oxides. The above results have indicated that $MgSiO_3$ was a remarkable candidate for the new biocompatible piezoelectric material.

3.3 Crystal Growth Design

We focused on $MgSiO_3$ as a new lead-free piezoelectric material, which has the highest piezoelectricity of 5 discovered oxides except $BaTiO_3$ existing. In order to generate a tetragonal $MgSiO_3$ thin film, crystal growth on various substrates was evaluated through the DFT calculations.

Total energies of tetragonal $MgSiO_3$ under various mechanical strains were evaluated by using the DFT. The k-point was used to sample the Brillouin zone with $8 \times 8 \times 8$ meshes and 500 eV for the cutoff energy. Lattice constants of the stable tetragonal structure of $MgSiO_3$, as $a = b = 0.3449$ nm and $c = 0.3538$ nm as shown in Table 5, were employed.

Total energies of tetragonal $MgSiO_3$ were calculated under 40 strain conditions, which were determined by the LHS method within the range from -1.0 % to 1.0 %. These total energies were approximated by the KHPA method to obtain the quadric model. In order to understand the nature of the obtained approximation function for the total energy, Fig. 11 shows the relationship between the normal strains ε_a, ε_b and the total energy under the condition of zero shear strain. The normal strains ε_c were Fig. 11 (a) 0.0 %, (b) 1.0 % and (c) -1.0 %, respectively. The approximated total energy in-

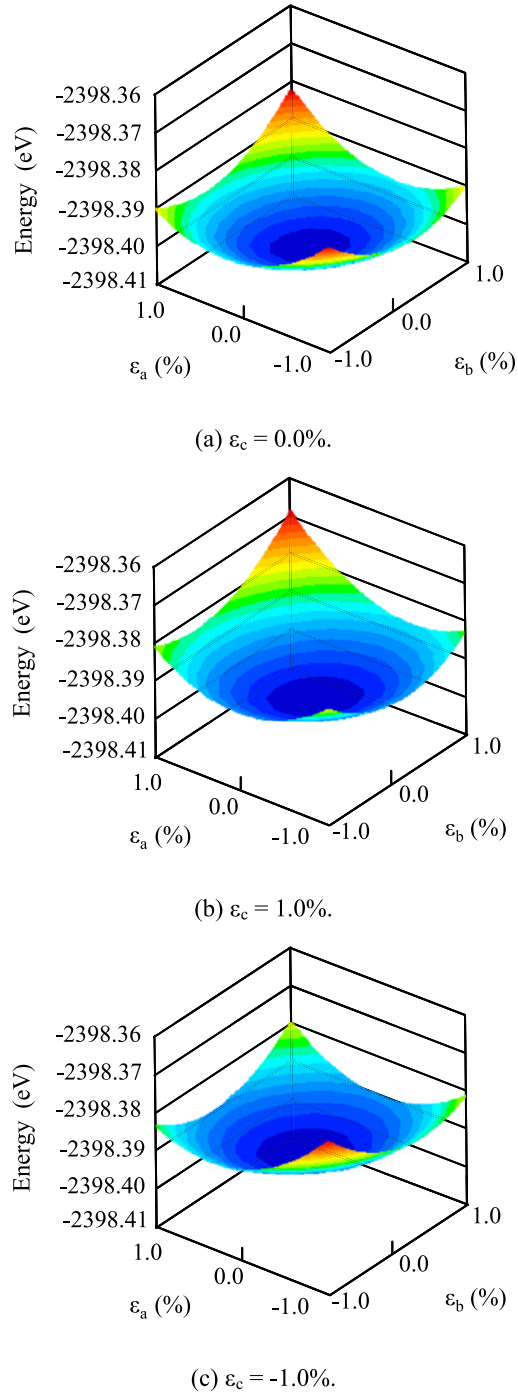


Fig. 11 Energy evaluation of tetragonal $MgSiO_3$ by using the approximate energy function of crystal unit cell with normal strains

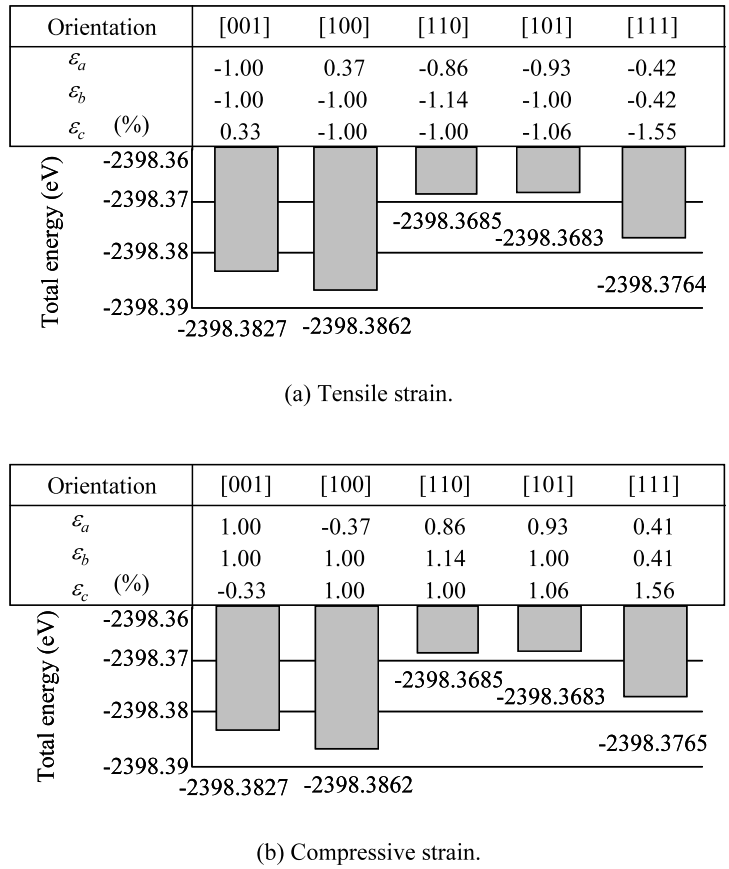


Fig. 12 Comparison of total energy among different crystal orientations under the same tensile and compressive epitaxial strain

dicated a minimum value at the zero strain state. In the case of $\varepsilon_c > 0$, the total energy of the crystal with compressive ε_a and ε_b was smaller than one with tensile ε_a and ε_b. In contrast, when $\varepsilon_c < 0$, the total energy of the crystal with tensile ε_a and ε_b was smaller than one with compressive ε_a and ε_b.

Figure 12 shows the comparison of total energies in the growth directions under the condition of 1.0 % epitaxial tensile and compression strains caused by the lattice mismatch with the substrate. In both cases of tensile and compressive strains, calculated result shows that if [001] and [100] orientations crystals grow on the substrate, the total energy becomes lower than other orientations. It indicates that these orientations had a high crystal growth possibility. In contrast, [110] and [101] orientations have a high total energy and are difficult to grow on the substrate. From this calculated result, we can predict the preferred orientations from the difference of the growth potential evaluated by using the first-principles calculation under the condition of the epitaxial strain with the lattice mismatch.

Four biocompatible atoms, which include Au, Mo, Fe and In, were selected for the substrate candidate. This is because:

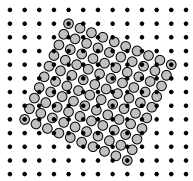
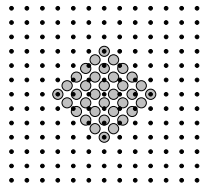
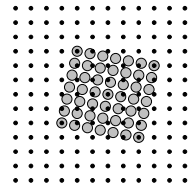
1. these atoms can be used for the under electrode.
2. these atoms have the cubic or the tetragonal crystal structures.

Lattice constants of Au with FCC cubic structure are $a = b = c = 0.4080$ nm, and ones of Mo with BCC cubic structure are $a = b = c = 0.3147$ nm. Ones of Fe with BCC cubic structure are $a = b = c = 0.2690$ nm. Additionally, ones of In with the tetragonal structure are $a = b = 0.3250$ nm and $c = 0.4950$ nm.

Crystal growth process of $MgSiO_3$ thin film on (100) and (111) facets of candidate substrates were demonstrated by using the process crystallographic simulation. Table 7 show an example of numerical results for $MgSiO_3$ thin film grown on Au(100) substrate. This table includes the conformation, mechanical strain in crystallographic coordinate system caused by lattice mismatch, total energy and total energy increment compared with unstrained unit cell.

Table 8 shows summary of the orientation fractions of $MgSiO_3$ thin film on substrates calculated by canonical distribution. In the case of Mo(100) substrate, $MgSiO_3$[100] and [001] were grown, and their orientation fraction were 61.5 % and 38.5 %, respectively. $MgSiO_3$[001] was grown on Au(100), Fe(100) and In(100) substrates at 100 % probability. Comparing total energy increments of crystal clusters of $MgSiO_3$ grown on these three substrates, $MgSiO_3$[001] on In(100) substrate was most stable due to the lowest total energy increment. $MgSiO_3$[111] was grown on Au(111) and Mo(111) substrates at 100 % provability. Comparing to-

Table 7 Analytical results for stable conformations and preferred orientations of $MgSiO_3$ thin film grown on Au(100) substrate

Orientation	[001]	[001]	[001]
Conformation			
ε_a (%)	0.21	0.48	0.64
ε_b (%)	0.21	0.48	0.64
ε_c (%)	0.00	0.00	0.00
γ_{ab} (%)	0.00	0.00	0.00
γ_{bc} (%)	0.00	0.00	0.00
γ_{ca} (%)	0.00	0.00	0.00
Total energy of the unit cell (eV)	−2398.4025	−2398.3984	−2398.3946
Total energy increment (eV)	0.0749	0.1257	0.3158

tal energy increments of crystal clusters of $MgSiO_3$ grown on these two substrates, Au(111) was better substrate than Mo(111) due to low total energy increment. $MgSiO_3$[111] and [001] on Fe(111) were grown at 97.8 % and 2.2 % probability, respectively. $MgSiO_3$[110] was grown on In(111) substrate at 100 %. Consequently, it could be concluded that five substrates, which included Mo(100), In(100) and(111), Au(111) and Fe(111), were candidates of the best substrate for $MgSiO_3$ thin film.

3.4 Material Property Evaluation

Piezoelectric stress constants of $MgSiO_3$ crystal were $e_{33} = 4.57$ C/m^2, $e_{31} = -2.20$ C/m^2 and $e_{15} = 12.77$ C/m^2, respectively, as shown in Table 6. Analytically determined piezoelectric stress constants and orientation fractions of $MgSiO_3$ were introduced into the macro continuum structure analysis. Homogenized piezoelectric strain constants of the $MgSiO_3$ thin film on five substrate candidates were calculated.

Figures 13 and 14 show homogenized e_{31} and e_{33} constants, respectively. Substrates, facets of substrates and orientation fractions of $MgSiO_3$ thin film were also shown in figures. $MgSiO_3$[111] on Au(111) substrate indicated the highest piezoelectric stress constants, $e_{31} = -3.65$ C/m^2 and $e_{33} = 5.10$ C/m^2. While, $MgSiO_3$[110] on In(111) did not have piezoelectric strain constants. This is because spontaneous polarization directed horizontally for substrate surface and $MgSiO_3$ was fixed. Therefore, it could not generate stress by applying vertical voltage. $MgSiO_3$[001] on In(100) showed $e_{31} = -2.20$ C/m^2 and $e_{33} = 4.57$ C/m^2. e_{31} of $MgSiO_3$[001] on In(100) was 39.7 % lower than one on Au(111) and e_{33} of $MgSiO_3$[001] on In(100) was 10.4 % lower than one on Au(111). In the case of Fe(111) substrate, e_{33} was equal to one on Au(111) substrate, however e_{31} was smaller than one on Au(111). Furthermore, $MgSiO_3$ on Au(111) was more stable than one on Fe(111) substrate. These results have concluded that Au(111) was the best substrate for $MgSiO_3$ thin film.

Table 8 Analytical results of preferred orientations and their fractions for $MgSiO_3$ thin films grown on various substrates

Substrate		$MgSiO_3$	
Atom	Facet	Orientation	Fraction (%)
Au	(100)	[001]	100.0
	(111)	[111]	100.0
Mo	(100)	[100]	61.5
		[001]	38.5
	(111)	[111]	100.0
Fe	(100)	[001]	100.0
	(111)	[111]	97.8
		[001]	2.2
In	(100)	[001]	100.0
	(111)	[110]	100.0

4 Thin Film Generation and Numerical Method Validation

This chapter describes the validation of newly proposed CAE simulation technique through comparison with experimental observations as "physical truth confirmation." We generated thin film of newly discovered biocompatible

Table 9 Classification of optimization technique

Optimization	Method	Characteristics
Functional analysis	Newton & Leibniz gradient method Lagrange multiplier method	Weak for practical use
Mathematical programing	Linear programming Nonlinear programming (gradient method: steepest decent method, conjugate gradient method)	Objective function formulated explicitly, time consuming
Numerical simulation and discrete variable optimization	Neural network Genetic algorithm Experimental design method Response surface method	Effective method for multi-design variable and multi-objective function

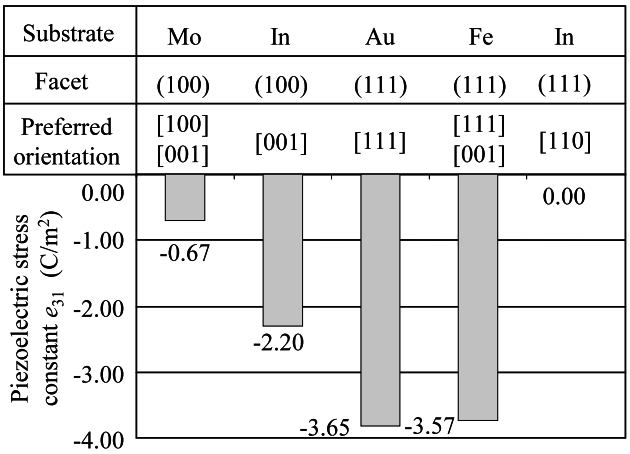


Fig. 13 Homogenized piezoelectric stress constant e_{31} of $MgSiO_3$ thin film on various substrates

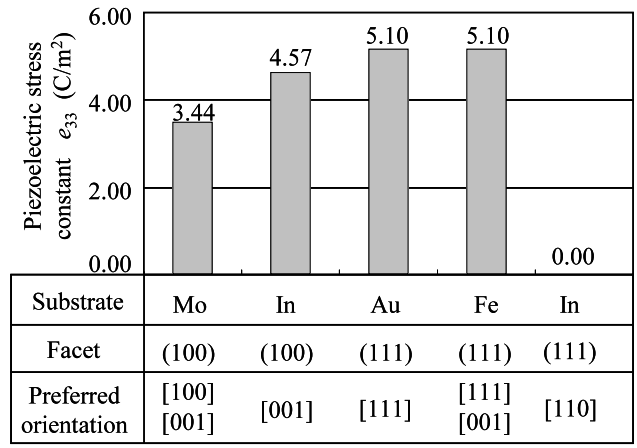


Fig. 14 Homogenized piezoelectric stress constant e_{33} of $MgSiO_3$ thin film on various substrates

piezoelectric material $MgSiO_3$ by RF magnetron sputtering apparatus, where we adopted the process optimization technique, such as the experimental design method—Taguchi method—and the surface response method (RSM) [99–108]. These are effective optimization algorithms for the problem of highly nonlinear and implicit objective function, and discrete design variable. Furthermore we confirm the overall CAE simulation's availability by comparison with characteristics measurement of generated $MgSiO_3$ piezoelectric thin film.

Generally, the enstatite ($Mg_2Si_2O_6$) crystal with perovskite type orthorhombic structure exists in the mantle of the Earth, and it does not have the piezoelectric property [109–112]. Ito et al. [113] fabricated $MgSiO_3$ with orthorhombic structure by the sintering method. Tanaka et al. [114] generated the $MgSiO_3$ with orthorhombic structure by using the top-seeded solution growth (TSSG) method. Pacalo et al. [115] generated $MgSiO_3$ with tetragonal structure and measured the elastic moduli of single crystal. However, $MgSiO_3$ material with the piezoelectric property has not been generated artificially yet.

In this paper, the crystallographic orientation and the piezoelectric property of the $MgSiO_3$ thin film were measured by the X-ray diffractometer (XRD) (X'pert: PANalytical) and the ferroelectric character evaluation (FCE) system (Nano-R: Toyo Corporation), respectively. These values are used as the objective functions. The dominant process variables were found by using the experimental design method [116]. At the final stage of process optimization, we employed the surface response method (RSM) to generate $MgSiO_3$ thin film with highest piezoelectric property. The cytotoxicity test was also conducted to confirm the biocompatibility of the $MgSiO_3$ thin film.

4.1 Optimization Method for Thin Film Generation Process

(A) Methodologies Many algorithms have been developed for various kinds of optimization problems as indicated in Table 9 [99–108]. The experimental and computational approaches can be quite time consuming. Further, in this thin film generation problems, only, the discrete optimization algorithm by using the numerical simulation technique is

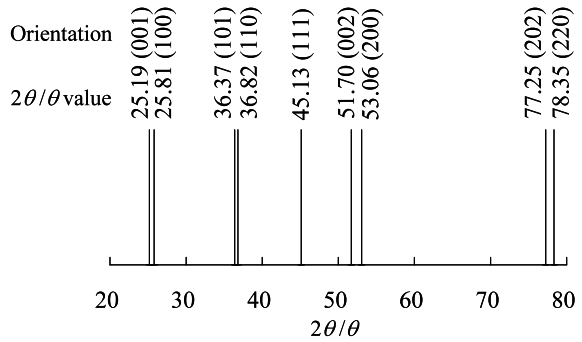


Fig. 15 $2\theta/\theta$ pattern of $MgSiO_3$ thin film calculated by first-principles DFT

Table 10 Analytical results of preferred orientations and their fraction for BTO thin films generated on various substrates

Substrate	Orientation of BTO	Fraction (%)
$SrTiO_3$(100)	[001]	58.5
	[100]	41.5
$SrTiO_3$(110)	[110]	94.7
	[001]	4.7
	[100]	0.6
$SrTiO_3$(111)	[111]	100.0
MgO(100)	[100]	53.6
	[001]	46.4

available, because we can't formulate the objective function by using design variables, explicitly. Therefore, numerical simulations, such as experimental design method [99–106] and the response surface method [107, 108], which is based on the robust modeling and effective calculation schemes, are employed.

(B) Design Variables Eight process design variables—generation conditions—of RF magnetron sputtering were determined as follow; (i) the electric power, (ii) the pressure of argon and oxygen atmosphere, (iii) the sputtering time, (iv) the flow rate of argon gas, (v) the flow rate of oxygen gas, (vi) the substrate temperature, (vii) the buffer layer and (viii) the post-annealing temperature. They are assigned to the orthogonal array in the experimental design algorithm [99] and the extended RSM calculation [107, 108].

(C) Objective Functions Two objective functions were employed, (1) Intensity of $MgSiO_3$ (111) crystal orientation. XRD $2\theta/\theta$ pattern of $MgSiO_3$ was calculated using Bragg's law by employing analytically determined lattice constants.

$$n\lambda = 2d\sin\theta, \tag{66}$$

$$\frac{1}{d_l^2} = \frac{h^2}{a^2} + \frac{k^2}{b^2} + \frac{l^2}{c^2}, \tag{67}$$

where, $\lambda(=0.1541$ nm) is the wavelength of X-ray by using Cu K_α radiation, d_l lattice distance between two adjacent planes, θ diffraction angle, h, k and l Miller indices, and a, b and c lattice constants. $2\theta/\theta$ of $MgSiO_3$ (111) was evaluated as 45.13 as shown in Fig. 15. (2) Piezoelectric constant d_{33}. The displacement-voltage curve of the $MgSiO_3$ thin film was measured by using the FCE system. The piezoelectric strain constant d_{33} was calculated by using gradient at the cross point of this butterfly-type hysteresis curve.

Finally, an optimum generating condition of the $MgSiO_3$ thin film, which showed the highest d_{33}, was determined.

4.2 Process Optimization of $MgSiO_3$ thin film generation

Before a generation of a new biocompatible piezoelectric material $MgSiO_3$, we verify this process crystallographic simulation technique by employing the existing material BTO ($BaTiO_3$). Following procedures were introduced to validate the process crystallographic simulation scheme.

1. The total energy function of BTO as a function of strain was derived by using the first-principles DFT. The characteristic of total energy of $BaTiO_3$ was derived and the validity of the function was confirmed.
2. The crystal growth process of BTO was analyzed by using the process crystallographic simulation. Preferred

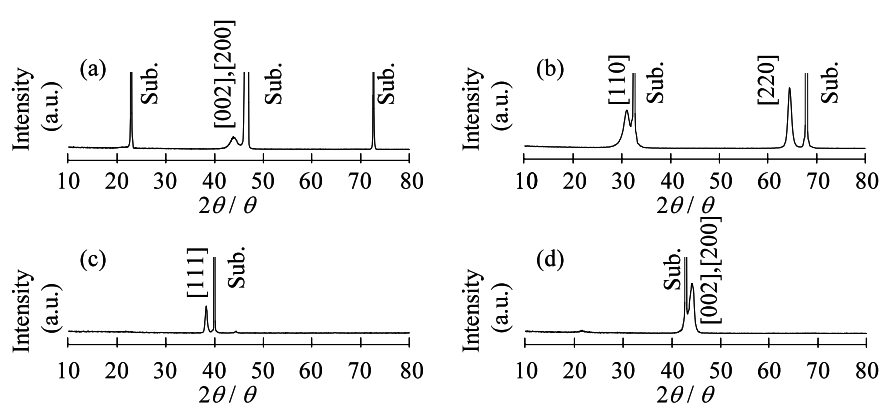


Fig. 16 XRD patterns for BTO thin films deposited on (**a**) $SrTiO_3$(100), (**b**) $SrTiO_3$(110), (**c**) $SrTiO_3$(111) and (**d**) MgO(100)

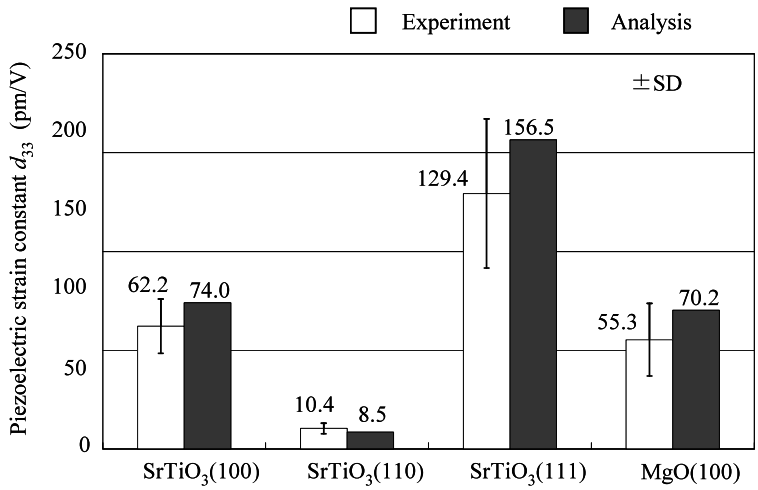


Fig. 17 Comparison of d_{33} between numerical simulation and experimental results

orientations of BTO piezoelectric thin film and their fractions were also calculated.

3. Macro homogenized properties were obtained to indicate characteristic of BTO thin film by using the macro continuum structure analysis based on the crystallographic homogenization method.
4. BTO thin films were generated by using the radio-frequency (RF) magnetron sputtering system. Its crystallographic orientations and piezoelectric properties were compared with analytically determined.

First, numerical results were obtained by using the first principle calculation, where the lattice constants were $a = b = 0.3932$ nm and $c = 0.3972$ nm, which had -1.55 % and -1.63 % errors by comparison with the experimentally measured values, as $a = b = 0.3992$ nm and $c = 0.4035$ nm, reported by Jaffe et al. [117].

Second, an epitaxial growth was simulated by employing four substrates, such as $SrTiO_3$(100), (110) and (111) and MgO(100), and we found the fraction of BTO crystal orientations, as shown in Table 10.

Third, the crystallographic homogenization simulations were executed to evaluate piezoelectric constants, d_{31} and d_{33}, in cases of BTO poly-crystal structures.

Fourth, we generated BTO thin films, which were grown on $SrTiO_3$(100), (110) and (111) and MgO(100) substrates, by using the RF magnetron sputtering system (E-200S: Canon ANELVA Corporation). Figure 16 shows XRD analyses results of BTO in four cases.

Fifth, piezoelectric properties of the BTO thin films were measured by using the ferroelectric character evaluation (FCE) system (Nano-R: Toyo Corporation). Figure 17 shows a comparison of d_{33} with experimental results. Analytical result of d_{33} showed errors with a range of 18.3 % to 26.9 %. This is because the effect of the temperature could not be calculated in the DFT. Therefore, calculated lattice constants were generally smaller than the experimentally observed lattice constants. Additionally, macro homogenized piezoelectric strain constants were calculated by employing a non-strained single crystal property. Piezoelectric properties were affected by the strain, which was caused by the lattice mismatch with the substrate crystal. These reasons caused the error of the piezoelectric property. However, the three-scale structure analysis could provide the crystallographic orientation and the piezoelectric property with enough accuracy. It was concluded that the process crystal-

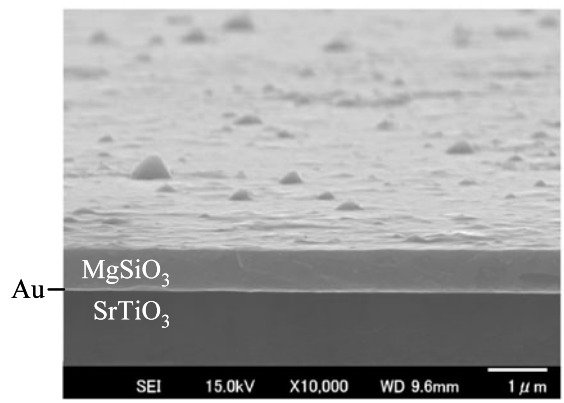


Fig. 18 SEM image of $MgSiO_3$ thin film in the case of $T_s = 300$ °C, $T_a = 650$ °C and $f_{O_2} = 3.0$ sccm

Table 11 Sputtering condition for the $MgSiO_3$ thin film

Target	Mixed sinter of MgO and SiO_2
Substrate	$SrTiO_3$(110)
Buffer layer	Au
Electric power	100 W
Flow rate of Ar gas	10 sccm
Pressure	0.5 Pa
Sputtering time	4 h
Post annealing time	1 h

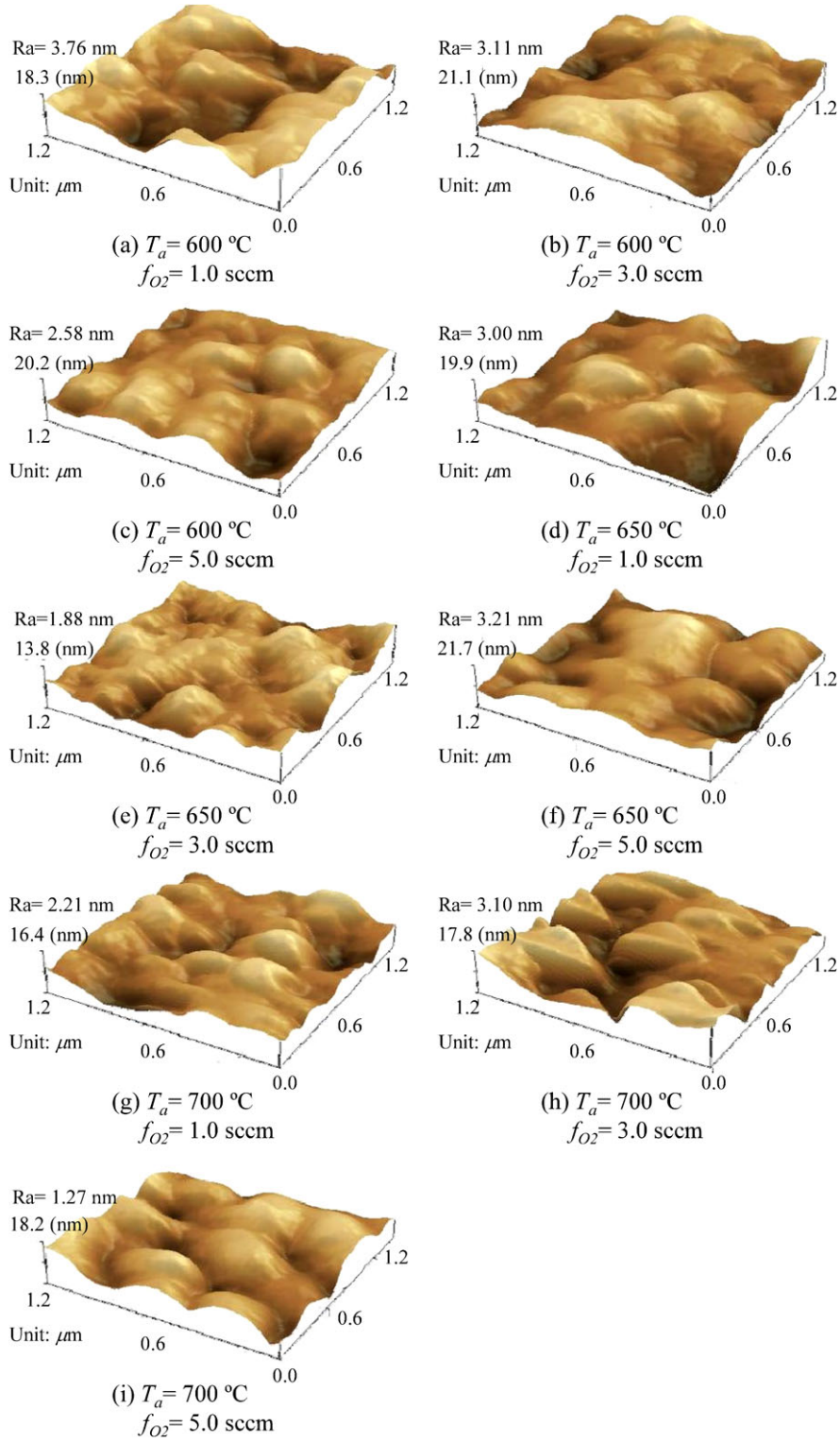


Fig. 19 AFM images of surfaces of $MgSiO_3$ thin films generated in cases of $T_s = 350$ °C

lographic simulation was a useful numerical tool to predict the crystal growth and the piezoelectric properties of the thin film.

After a validation of the process crystallographic simulation scheme, we applied to a new biocompatible piezoelectric $MgSiO_3$ thin film generation. The discovery of $MgSiO_3$ was already discussed in Sect. 3. At first, we adopted the heuristically determined generating conditions of RF magnetron sputtering as shown in Table 11. $MgSiO_3$ thin film was generated on Au doped $SrTiO_3$ substrate, which was

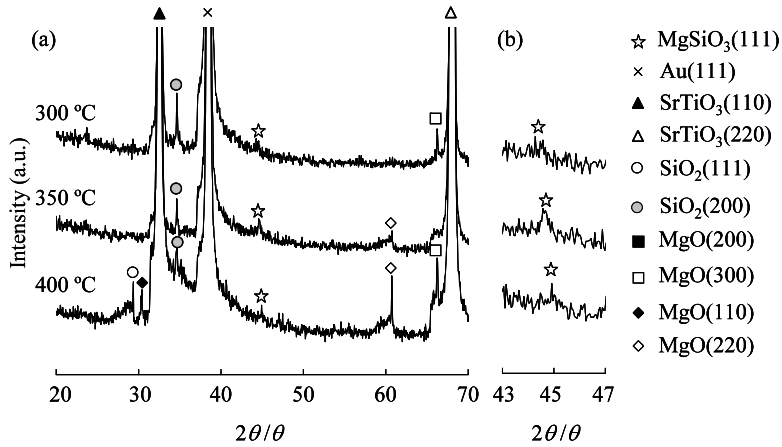


Fig. 20 XRD $2\theta/\theta$ patterns for $MgSiO_3$ thin film in the case of $T_a = 650$ °C and $f_{O_2} = 3.0$ sccm

Table 12 Three factors and three levels of $MgSiO_3$ thin film generation condition in the experimental design method

Substrate temperature T_s (°C)	300	350	400
Post annealing temperature T_a (°C)	600	650	700
Flow rate of oxygen gas f_{O_2} (sccm)	1.0	3.0	5.0

Table 13 The analysis of variance table (ANOVA table)

	S	ϕ	V	F_0
Factor	S_{Factor}	ϕ_{Factor}	$V_{Factor} = S_{Factor}/\phi_{Factor}$	V_{Factor}/V_{Error}
⋮	⋮	⋮	⋮	⋮
Error	S_{Error}	ϕ_{Error}	$V_{Error} = S_{Error}/\phi_{Error}$	
Total	S_T	ϕ_T		

Table 14 Analysis of variance table of the generation conditions for the $MgSiO_3$ thin film

Source	S	ϕ	V	F_0
T_s	5.76×10^4	2	2.88×10^4	7.37
T_a	5.67×10^3	2	2.84×10^3	0.73
f_{O_2}	9.48×10^3	2	4.74×10^3	1.21
$T_s \times T_a$	7.50×10^4	4	1.88×10^4	4.80
$T_s \times f_{O_2}$	1.23×10^5	4	3.07×10^4	7.87
$T_a \times f_{O_2}$	7.68×10^4	4	1.92×10^4	4.91
Error	2.04×10^6	521	3.91×10^3	
Total	2.38×10^6	539		

used for the lower electrode. Three design variables, (1) The substrate temperature T_s, (2) the post-annealing temperature T_a and (3) the flow rate of O_2 f_{O_2} were employed in the experiment design calculation [99, 100]. We searched the significant generation factor by employing three factors and three levels, as shown in Table 12, and "L27(3^{13}) array."

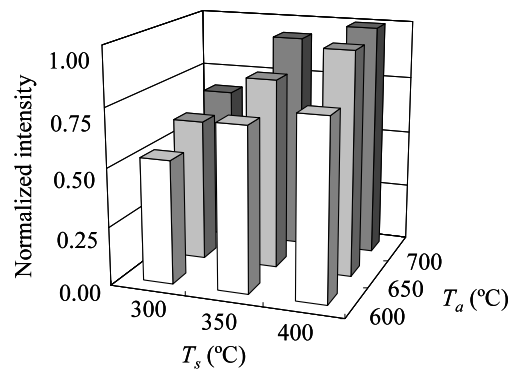


Fig. 21 Normalized X-ray intensities of $MgSiO_3$(111) peaks

A SEM image of $MgSiO_3$ thin film, which was epitaxially grown on Au/$SrTiO_3$/Si substrate, in the case of $T_s = 300$ °C, $T_a = 650$ °C and $f_{O_2} = 3.0$ sccm, is shown in Fig. 18. Its thickness was 750 nm.

AFM images of surfaces of $MgSiO_3$ thin films in the case of $T_s = 350$ °C are shown in Figs. 19(a)–(i).

XRD $2\theta/\theta$ patterns for $MgSiO_3$ thin film in the case of $T_a = 650$ °C and $f_{O_2} = 3.0$ sccm in the range between 20 and 70 of $2\theta/\theta$ values are shown in Fig. 20(a). Figure 20(b) enlarged view range from 43 to 47. $MgSiO_3$ crystal with the perovskite and tetragonal structure is an unstable crystal, and its components, SiO_2 and MgO, are more stable than it. If the energy supply was low, SiO_2 and MgO crystals were separately grown on the substrate, and then could not be crystallized to $MgSiO_3$. We have supplied enough energy for crystallization, therefore, peaks of $MgSiO_3$[111] crystal were obtained in all sputtering conditions. $MgSiO_3$[111] orientation intensity was divided by summation of SiO_2, MgO and $MgSiO_3$ orientations intensities to obtained the normalized value for the crystallinity evaluation of $MgSiO_3$. Figure 21 shows a comparison with intensities of $MgSiO_3$ [111] by using the full width at half maximum (FWHM) and the mean value of f_{O_2}. It shows that the higher the tem-

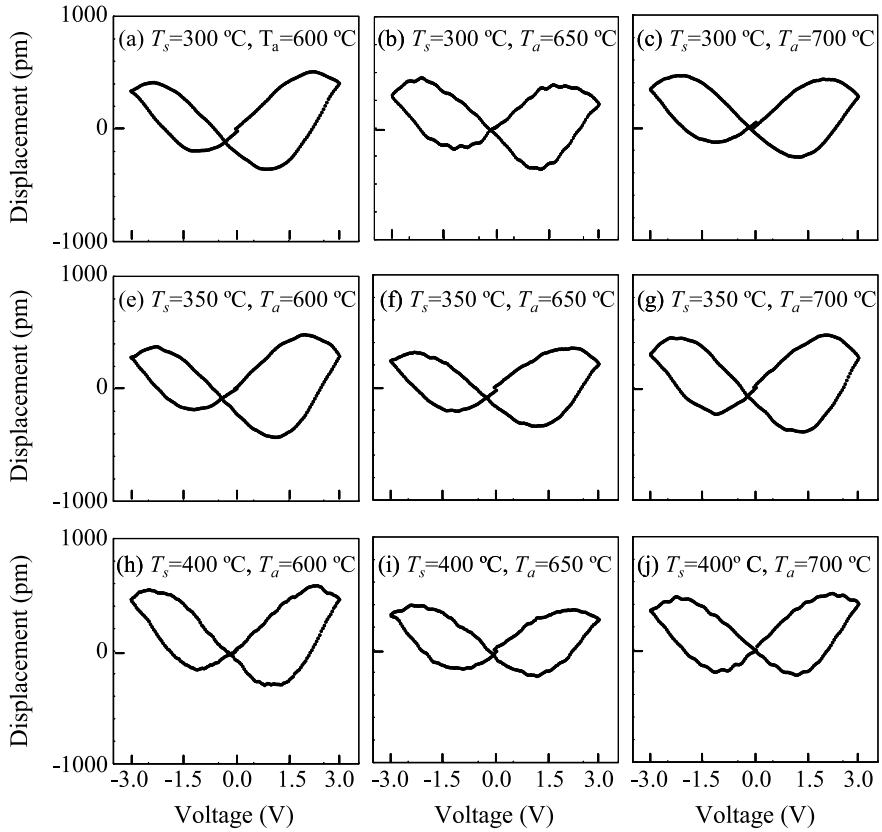


Fig. 22 Displacement-voltage curves of $MgSiO_3$ thin films in the case of $f_{O_2} = 3.0$ sccm

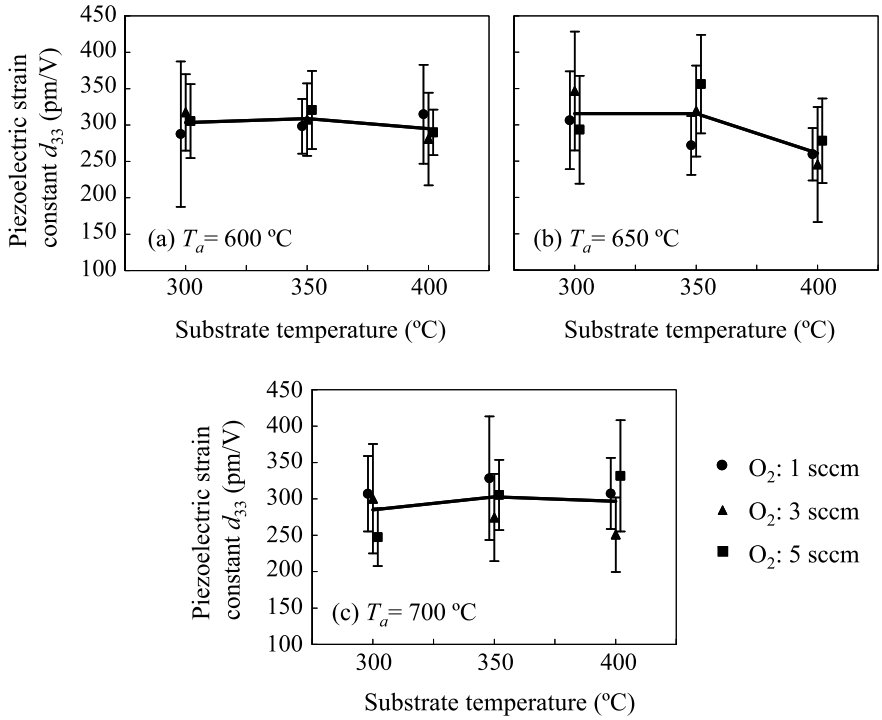


Fig. 23 Relationships between piezoelectric strain constant d_{33} and the substrate temperature under post-annealing temperatures of (**a**) 600 °C, (**b**) 650 °C and (**c**) 700 °C

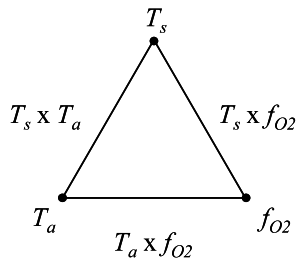


Fig. 24 Linear graph of the L27(3^{13})

peratures of the substrate during the epitaxial growth and the post-annealing, the higher the crystallization of $MgSiO_3$.

Figures 22(a)–(j) show displacement-voltage curves under the condition of $f_{O_2} = 3.0$ sccm, respectively. All thin films showed the piezoelectric property due to butterfly-type hysteresis curves.

Figures 23(a)–(c) show the relationship between d_{33} and the substrate temperature T_s in cases of (a) $T_a = 600$ °C, (b) 650 °C and (c) 700 °C, respectively. Lines in figures show the average d_{33} value at each post-annealing temperature.

The efficient of main effects and their interactions for piezoelectric strain constant d_{33} of the $MgSiO_3$ thin film were analyzed by the experimental planning method. L27(3^{13}) array was employed to calculate the efficient as shown in Fig. 24. The degree of freedom ϕ, the variation S, the variance V and the variance ratio F_0 were summarized in an analysis of variance (ANOVA) table as shown in Table 13 to proceed to F-testing. Table 14 summarizes three three-level factors of T_s, T_a and f_{O_2} and three two-factor interactions $T_s \times T_a$, $T_s \times f_{O_2}$ and $T_a \times f_{O_2}$. These results have indicated that the substrate temperature was significant at 1 % level and all interactions were also significant at 1 % level. On the other hand, the post-annealing temperature and the flow rate of oxygen were not significant. It indicated that the post-annealing temperature T_a was higher than the crystallization temperature T_s the flow rate of oxygen f_{O_2} can be neglected, if we adopted the value more than 3.0 sccm, suc as $f_{O_2} \geq 3.0$ sccm.

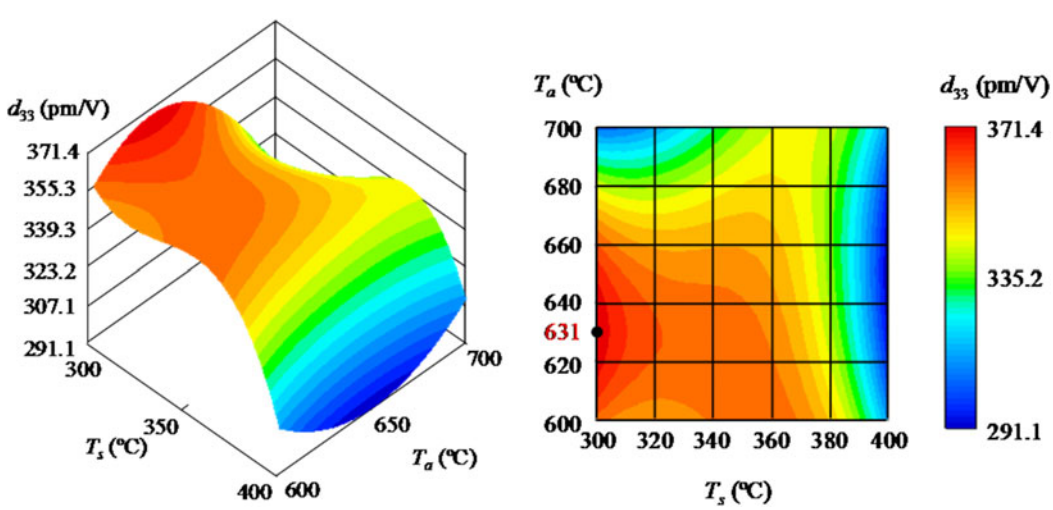
At the final stage of process optimization, RSM calculation was executed. The optimum conditions for generating the $MgSiO_3$ thin film were found by using RSM. Figures 25(a) and (b) show the response surfaces of d_{33} under the condition of $f_{O_2} = 4.0$ sccm as a function of T_s and T_a in cases of aerial and top views. The optimum condition, at $T_s = 300$ °C and $T_a = 631$ °C, was found.

4.3 Validation of CAE Driven Material Design and Generation—Piezoelectricity and Biocompatibility

In this chapter, the validation of the "process crystallographic" simulation technique was demonstrated. The piezoelectric property and biocompatibility were examined to confirm CAE driven new material $MgSiO_3$ development [116, 118–120].

(A) Piezoelectricity $MgSiO_3$ thin film was generated at the optimum condition obtained in Sect. 4.2. Figure 26 shows displacement-voltage curve. d_{33} constant was obtained as 359.2 pm/V. This value is compatible to one of PZT thin film, which was generated by Zhu et al. [120].

(B) Biocompatibility The biocompatibility of $MgSiO_3$ was examined through the cytotoxicity test [118, 119] by using the L929 cell in Dulbecco's modified Eagle medium (DMEM) supplemented with 10 % fetal bovine serum



(a) 3D view (b) Top view

Fig. 25 RMS of Piezoelectric strain constant d_{33} as functions of T_s and T_a in the case of $f_{O_2} = 4.0$ sccm

(FBS). Extraction liquids of five piezoelectric material samples, $MgSiO_3$, $CaSiO_3$, $CaTiO_3$, BTO and PZT, and two standard samples, such as the toxic material (Ref. A)—the polyurethane film with 0.25 % zinc dibutyldithio carbamate—and the non-toxic material (Ref. B) were employed. The cells cultured in the medium were called as a "control." The cell number cultured in the extraction liquid of the biocompatible sample should be higher than one of the Ref. A. The cells were seeded on six-well plates with DMEM supplemented with 10 % FBS at cell density of 6.67×10^3 cells/ml. They were incubated at 37 °C, in a 100 % humidified atmosphere containing 5 % carbon dioxide gas.

The cell doubling time was calculated by following equation.

$$T_{double} = \frac{(t_i - t_0)\log 2}{\log N_{cell\ i} - \log N_{cell\ 0}}, \tag{68}$$

where t_0 is the standard time and $N_{cell\ 0}$ the cell number at t_0.

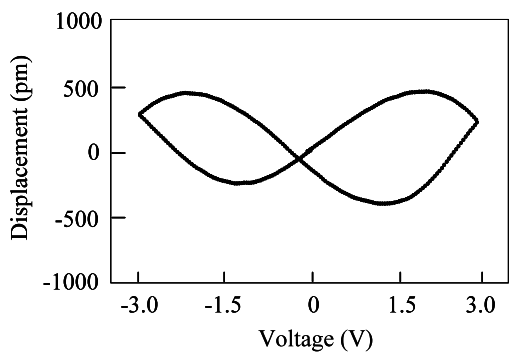


Fig. 26 Displacement-voltage curve of $MgSiO_3$ thin film generated at the optimum condition, $T_s = 300$ °C, $T_a = 631$ °C and $f_{O_2} = 4.0$ sccm

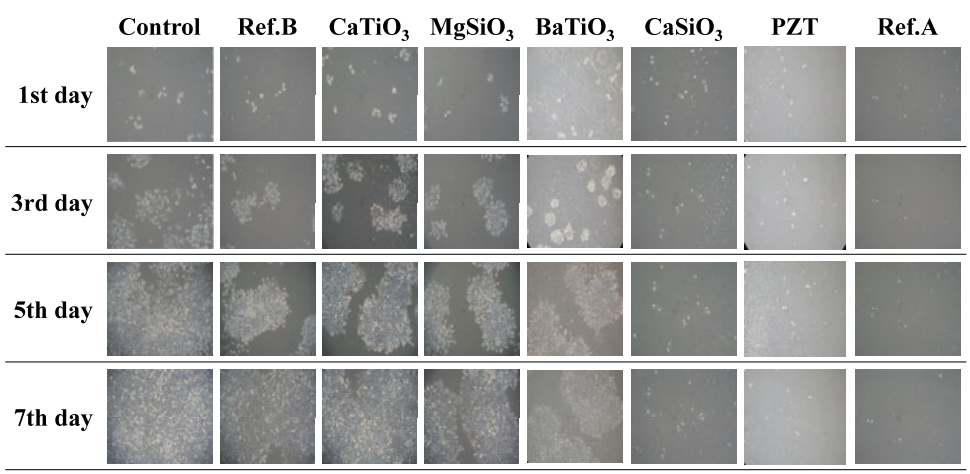
Microscopic observation images of cell proliferation during the cytotoxicity test are shown in Fig. 27. The cells of "$MgSiO_3$" increased as the same as those of the control. However, PZT killed cells and shows the similar configuration of toxic material Ref. A. The doubling time is shown in Fig. 27. These results show that $MgSiO_3$ has a high biocompatibility.

Consequently, the piezoelectricity and biocompatibility of $MgSiO_3$ thin film, which were designed by "process crystallographic" simulation and generated through the process optimization by using the experimental design method and RSM, were confirmed.

5 Conclusion

This paper describes an overview of "process crystallographic simulation technique" to solve the industrial requirement for a new biocompatible piezoelectric material design and its generation process optimization. This CAE driven material design and generation technique consists of (1) three scale analysis: the first principle calculation for crystal structure design and biocompatible element selection, the two-scale finite element analysis based on the crystallographic homogenization theory, and (2) thin film generation process optimization by using sputtering machine. The selection of substrate for epitaxial growth was performed by geometrical approximation theory. Finally, availability of CAE driven technique was demonstrated by comparison with experimentally generated thin film.

Fig. 27 Microscopic observation images of cell proliferation during the cytotoxicity test and the doubling time table



	Control	$CaTiO_3$	Ref. B	$MgSiO_3$	$BaTiO_3$	$CaSiO_3$	PZT	Ref. A
Doubling Time (hr)	20.15	21.45	22.96	23.89	27.11	NA	NA	NA

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